Mathematics II

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You can find the last version of these course materials at

https://mbujosab.github.io/MatematicasII/


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Contents

VI Eigenvalues and eigenvectors **1**

LECTURE 16: Eigenvalues and eigenvectors **2**
Slides for Lecture 16 . . . 2
Questions of the Lecture 16 . . . 6

LECTURE 17: Diagonalization in triangular blocks by similarity **9**
Slides for Lecture 17 . . . 9
Questions of the Lecture 17 . . . 13

LECTURE 18: Symmetric Matrices and Orthogonal Diagonalization **16**
Slides for Lecture 18 . . . 16
Questions of the Lecture 18 . . . 18

LECTURE 19: Diagonalization by congruence. Definite Matrices **21**
Slides for Lecture 19 . . . 21
Questions of the Lecture 19 . . . 26

OPTIONAL LECTURE II: Review execises **30**
Questions of the Optional Lecture 2 . . . 30

Solutions **33**

Part VI
Eigenvalues and eigenvectors

LECTURE 16: Eigenvalues and eigenvectors

Lecture 16

(**Lecture** 16) **S-1** **Overview of Lesson 16**

Always squared matrices in this topic

Overview of *Lesson 16*

- **Eigenvalues, eigenvectors** (the prefix eigen is the German word for innate, distinct, self)
- $|\mathbf{A} - \lambda \mathbf{I}| = 0$ *Characteristic equation*
- $\mathrm{tr}\,(\mathbf{A})$, $\det \mathbf{A}$ (demo in the next lesson)

F1

(**Lecture** 16) **S-2** **Eigenvalues and eigenvectors**

Consider the equation

$$\mathbf{A}\boldsymbol{x} = \lambda \boldsymbol{x} \qquad (\text{with} \quad \boldsymbol{x} \neq \mathbf{0})$$

- *Eigenvalue* is any λ such that there are solutions.
- Such *non-zero* solutions $\boldsymbol{x}$ are called *eigenvectors*.

$\boldsymbol{x} \neq \mathbf{0}$ such that $\mathbf{A}\boldsymbol{x}$ is a multiple of $\boldsymbol{x}$

When λ is 0, what are the eigenvectors?

.

F2

(**Lecture** 16) **S-3** **Example: projection matrix**

- Orthogonal projection
- Which vectors are eigenvectors?
  What vectors remain in the same direction?
- What are the eigenvalues of those eigenvectors?
- Are there any other eigenvectors? With what eigenvalue?
- Two eigenspaces

F3

(**Lecture 16**) **S-4** **Another example: swap matrix**

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- A vector that does not change after the swap?
- What is the eigenvalue?
- Is there an eigenvector associated with $\lambda_2 = -1$?

$$\mathbf{A}\boldsymbol{x}_2 = -\boldsymbol{x}_2$$

Note: $\operatorname{tr}(\mathbf{A}) = 0 = \lambda_1 + \lambda_2$; $\det \mathbf{A} = -1 = \lambda_1 \cdot \lambda_2$. F4

The trace of an n-by-n square matrix $\mathbf{A}$ is defined to be the sum of the elements on the main diagonal:

$$\operatorname{tr}(\mathbf{A}) = a_{11} + a_{22} + \cdots + a_{nn}.$$

(**Lecture 16**) **S-5** **How to find eigenvalues and eigenvectors?**

How to solve

$$\mathbf{A}\boldsymbol{x} = \overbrace{\lambda}^{?}\overbrace{\boldsymbol{x}}^{?} \; ?$$

Let's rearrange ...

$$(\mathbf{A} - \lambda\mathbf{I})\boldsymbol{x} =$$

idea For this to hold, what kind of matrix must be $(\mathbf{A} - \lambda\mathbf{I})$?
and then what must its determinant equal? $|\mathbf{A} - \lambda\mathbf{I}| =$ F5

(**Lecture 16**) **S-6** **How to find eigenvalues and eigenvectors?**

1. Eigenvalues are the λ's such that: $|\mathbf{A} - \lambda\mathbf{I}| =$

   (Characteristic polynomial $P_{\mathbf{A}}(\lambda)$)

2. How to compute $\boldsymbol{x}$ such that $(\mathbf{A} - \lambda\mathbf{I})\,\boldsymbol{x} = \mathbf{0}$?

Eigenspace (Set of eigenvectors + $\mathbf{0}$):

$$\mathcal{E}_{\lambda}(\mathbf{A}) = \left\{ \boldsymbol{x} \in \mathbb{R}^n \,\middle|\, \mathbf{A}\boldsymbol{x} = \lambda\boldsymbol{x} \right\}$$

Spectrum: set $\{\lambda_1, \ldots, \lambda_k\}$ of eigenvalues $\big($roots of $P_{\mathbf{A}}(\lambda)\big)$ F6

(**Lecture** 16) S-7 **Example (we must compute the eigenvalues first!)**

We are looking for a null determinant (Characteristic polynomial)

$$\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}; \qquad \det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} = (3-\lambda)^2 - 1 = 0$$

Note: $\operatorname{tr}(\mathbf{A}) = 6 = \lambda_1 + \lambda_2$; $\det \mathbf{A} = 8 = \lambda_1 \cdot \lambda_2$.

F7

Remember that if $ax^2 + bx + c = 0$, then the value x that solves this quadratic equation is

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

See:
https://itempool.com/3b1b/c/ag9L0EqZFtK
http://en.wikipedia.org/wiki/Quadratic_equation

```
L = sympy.symbols('\lambda')
A = Matrix([[3,1],[1,3]])
D = Determinante(A-L*I(2),1).valor
display(D)

d = sympy.poly(D)
r = sympy.real_roots(d)
display(r)

ElimG(A-L*I(2), 1)
```

(**Lecture** 16) S-8 **Example (. . . and then the eigenspaces)**

And now we compute the null space $\mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$ . . . for each λ.

For $\lambda_1 = 4$

$$(\mathbf{A} - 4\mathbf{I}) = \begin{bmatrix} 3-4 & 1 \\ 1 & 3-4 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \Rightarrow$$

For $\lambda_2 = 2$

$$(\mathbf{A} - 2\mathbf{I}) = \begin{bmatrix} 3-2 & 1 \\ 1 & 3-2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \Rightarrow$$

Are they the only two eigenvectors?

$$\mathbf{A}\boldsymbol{x}_i = \lambda \boldsymbol{x}_i; \qquad \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \boldsymbol{x}_i = \lambda \boldsymbol{x}_i.$$

F8

(**Lecture** 16) **S-9** **Another example: 90º rotation matrix**

$$\mathbf{Q} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

- What is the sum of the eigenvalues?
- What is the determinant?

Difficulties

$\lambda_1 + \lambda_2 = 0$ and $\lambda_1 \cdot \lambda_2 = 1$ $\qquad (+)\cdot(-) = (+)$?

What kind of vector can be parallel to itself after a 90º rotation?

$$\det(\mathbf{Q} - \lambda \mathbf{I}) = \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 + 1 =$$

F9

$$\mathbf{Q}\boldsymbol{x} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

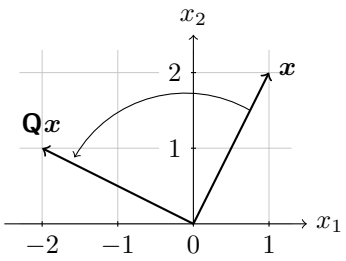


Figure 1: Rotation

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{pmatrix} i \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ i \end{pmatrix} = i \begin{pmatrix} i \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{pmatrix} 1 \\ i \end{pmatrix} = \begin{pmatrix} -i \\ 1 \end{pmatrix} = -i \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

Somehow these complex vectors keep their direction as they are rotated. Don't ask me how!

Symmetric matrices always have *real* eigenvalues (and perpendicular eigenvectors —as we will see soon), but *anti-symmetric* (like $\mathbf{Q}$) matrices have purely imaginary eigenvalues.

(Lecture 16) S-10 There are even worse examples

$$\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$$

- Eigenvalues

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 3-\lambda & 1 \\ 0 & 3-\lambda \end{vmatrix} = (3-\lambda)(3-\lambda) = 0 \begin{cases} \lambda_1 = 3 \\ \lambda_2 = 3 \end{cases}$$

- Eigenvectors
  - for λ_1: $(\mathbf{A} - \lambda \mathbf{I})\boldsymbol{x} = \mathbf{0} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \boldsymbol{x}_1$; $\qquad \boldsymbol{x}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
  - for λ_2:

$\lambda = 3$ is repeated twice, but $\dim \mathcal{E}_3(\mathbf{A}) = 1$

$$\mu(3) = 2 \neq 1 = \gamma(3)$$

F10

Summary:

1. The eigenvalues are those numbers λ that make the matrix $(\mathbf{A} - \lambda \mathbf{I})$ singular. In other words, they are the roots of the characteristic polynomial: $\det(\mathbf{A} - \lambda \mathbf{I})$.
2. Any $n \times n$ matrix has a characteristic polynomial of degree n
3. A polynomial of degree n has n roots (perhaps some roots are repeated).
4. The sum of the eigenvalues of a matrix equals its trace
5. The product of the eigenvalues of a matrix equals its determinant
6. The eigenvectors associated with λ are the non-zero vectors in $\mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$.

The lecture ends here

Questions of the Lecture 16

(L-16) Question 1. Consider the matrix

$$\mathbf{A} = \begin{bmatrix} -3 & 4 & -4 \\ -3 & 5 & -3 \\ -1 & 2 & 0 \end{bmatrix}$$

(a) The eigenvalues of $\mathbf{A}$ are -1, 1, and 2; and two corresponding eigenvectors are

$$\boldsymbol{v} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}; \qquad \boldsymbol{w} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

Verify that both vectors are indeed eigenvectors of $\mathbf{A}$. What are the corresponding eigenvalues?

(b) Find a third linearly independent eigenvector.

(L-16) Question 2. Find the eigenvalues and eigenvectors of

(a)

$$\mathbf{A} = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$$

(b)

$$\mathbf{B} = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$$

(Strang, 2006, exercise 12 from section 5.1.)

(L-16) Question 3. If $\mathbf{B}$ has eigenvalues 1, 2, 3, $\mathbf{C}$ has eigenvalues 4, 5, 6, and $\mathbf{D}$ has eigenvalues 7, 8, 9, what are the eigenvalues of the 6×6 matrix $\mathbf{A} = \begin{bmatrix} \mathbf{B} & \mathbf{C} \\ \mathbf{0} & \mathbf{D} \end{bmatrix}$? where $\mathbf{B}$, $\mathbf{C}$, $\mathbf{D}$ are upper triangular matrices.
(Strang, 2006, exercise 13 from section 5.1.)

(L-16) Question 4. Find the eigenvalues and eigenvectors of
(a)

$$\mathbf{A} = \begin{bmatrix} 3 & 4 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

(b)

$$\mathbf{B} = \begin{bmatrix} 0 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & 0 \end{bmatrix}$$

(Strang, 2006, exercise 5 from section 5.1.)

(L-16) Question 5. The eigenvalues of $\mathbf{A}$ are equal to the eigenvalues of $\mathbf{A}^\intercal$. This is because $\det(\mathbf{A} - \lambda\mathbf{I})$ is equal to $\det(\mathbf{A}^\intercal - \lambda\mathbf{I})$.
(a) This is true because _____
(b) Show with an example that, however, the eigenvectors of $\mathbf{A}$ and $\mathbf{A}^\intercal$ are not the same.
(Strang, 2006, exercise 11 from section 5.1.)

(L-16) Question 6. Let $\mathbf{B}$ be a matrix and $\boldsymbol{x}$ an eigenvector associated with the eigenvalue λ, that is $\mathbf{B}\boldsymbol{x} = \lambda\boldsymbol{x}$; and also consider the matrix $\mathbf{A} = (\mathbf{B} + \alpha\mathbf{I})$. Prove that $\boldsymbol{x}$ is also an eigenvector of $\mathbf{A}$ with associated eigenvalue $(\lambda + \alpha)$.

(L-16) Question 7.

(a) Encuentre los autovalores y los autovectores de la matriz $\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$. Compruebe que la traza es igual a la suma de los autovalores, y que el determinante es igual a su producto.
(b) Si consideramos una nueva matriz, generada a partir de la anterior como

$$\mathbf{B} = (\mathbf{A} - 7\mathbf{I}) = \begin{bmatrix} -6 & -1 \\ 2 & -3 \end{bmatrix}.$$

¿Cuáles son los autovalores y autovectores de la nueva matriz, y cómo están relacionados con los de $\mathbf{A}$?
(Strang, 2006, exercise 1 and 3 from section 5.1.)

(L-16) Question 8. Suppose that λ is an eigenvalue of $\mathbf{A}$, and that $\boldsymbol{x}$ is an eigenvector such that $\mathbf{A}\boldsymbol{x} = \lambda\boldsymbol{x}$.
(a) Show that the same $\boldsymbol{x}$ is also an eigenvector of $\mathbf{B} = \mathbf{A} - 7\mathbf{I}$, and find the corresponding eigenvalue of $\mathbf{B}$.
(b) Assume that $\lambda \neq 0$ (and that $\mathbf{A}$ is invertible), show that $\boldsymbol{x}$ is also an eigenvector of $\mathbf{A}^{-1}$, and find the corresponding eigenvalue. What is its relationship with λ?
(Strang, 2006, exercise 7 from section 5.1.)

(L-16) Question 9. Suppose that $\mathbf{A}$ is an $n \times n$ matrix, such that $\mathbf{A}^2 = \mathbf{A}$. What possible values can the eigenvalues of $\mathbf{A}$ take?

(L-16) Question 10. Suppose the matrix $\underset{3\times3}{\mathbf{A}}$ has eigenvalues 1, 2, and 3. If $\boldsymbol{v}_1$ is an eigenvector associated with the eigenvalue 1, $\boldsymbol{v}_2$ is associated with the eigenvalue 2, and $\boldsymbol{v}_3$ is associated with the eigenvalue 3; then what is $\mathbf{A}(\boldsymbol{v}_1 + \boldsymbol{v}_2 - \boldsymbol{v}_3)$?

(L-16) Question 11. Provide an example that shows that eigenvalues can change when a multiple of one column is subtracted from another. Why do elimination steps not change the zero eigenvalues?
(Strang, 2006, exercise 6 from section 5.1.)

(L-16) Question 12. The characteristic polynomial of a matrix $\mathbf{A}$ can be factored as

$$\det(\mathbf{A} - \lambda \mathbf{I}) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \cdots (\lambda_n - \lambda).$$

Show, starting from this factorization, that the determinant of $\mathbf{A}$ is equal to the product of its eigenvalues. To do this, make a smart choice of the value of λ.
(Strang, 2006, exercise 8 from section 5.1.)

(L-16) Question 13. Calculate the characteristic values (eigenvalues) and the characteristic vectors of $\mathbf{A}$ and $\mathbf{A}^2$:

$$\mathbf{A} = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix} \qquad \text{y} \qquad \mathbf{A}^2 = \begin{bmatrix} 7 & -3 \\ -2 & 6 \end{bmatrix}$$

$\mathbf{A}^2$ has the same _____ as $\mathbf{A}$. When the eigenvalues of $\mathbf{A}$ are λ_1 and λ_2, the eigenvalues of $\mathbf{A}^2$ are _____.
(Strang, 2006, exercise 22 from section 5.1.)

(L-16) Question 14. Assume that the characteristic values of $\underset{3\times 3}{\mathbf{A}}$ are 1, 2, and 4, what is the trace of $\mathbf{A}^2$? What is the determinant of $\left(\mathbf{A}^{-1}\right)^{\mathsf{T}}$?
(Strang, 2006, exercise 10 from section 5.2.)

(L-16) Question 15. The equation $(\mathbf{A}^2 - 4\mathbf{I})\boldsymbol{x} = \boldsymbol{b}$ has no solution for some right-hand side $\boldsymbol{b}$. Provide as precise information as possible about the eigenvalues of the matrix $\mathbf{A}$ (the matrix $\mathbf{A}$ is diagonalizable). *MIT Course 18.06 Quiz 3. Spring, 2009*

(L-16) Question 16. You are given the matrix

$$\mathbf{A} = \begin{bmatrix} 0.5 & 0.2 & 0.2 \\ 0.1 & 0.5 & 0.5 \\ 0.4 & 0.3 & 0.3 \end{bmatrix}$$

One of the eigenvalues is $\lambda = 1$. What are the eigenvalues of $\mathbf{A}$? [Hint: Very little calculation is required! You should be able to see another eigenvalue by inspecting the form of $\mathbf{A}$, and the third through a simple calculation. You should not need to calculate $\det(\mathbf{A} - \lambda \mathbf{I})$ unless you really want to do it the complicated way.] *MIT Course 18.06 Quiz 3. Spring, 2009*

End of Questions of the Lecture 16

LECTURE 17: Diagonalization in triangular blocks by similarity

Lecture 17

(**Lecture** 17) S-1 Highlights of Lesson 17

Highlights of *Lesson 17*

- Similar matrices: $\mathbf{C} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}$
- Triangular block diagonalization of a matrix

$$\begin{bmatrix}\mathbf{A}\\ \mathbf{I}\end{bmatrix} \xrightarrow[esp(\boldsymbol{\tau}_p^{-1}\cdots\boldsymbol{\tau}_1^{-1})]{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_p} \begin{bmatrix}\mathbf{C}\\ \mathbf{S}\end{bmatrix} \qquad \text{where} \quad \mathbf{S} = \mathbf{I}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_p}.$$

- Diagonalizable matrices: when $\mathbf{C}$ is diagonal.

F12

(**Lecture** 17) S-2 Similar matrices

$\mathbf{A}$ and $\mathbf{C}$ are *similar* if there exists an invertible $\mathbf{S}$ such that

$$\mathbf{C} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}$$

If $\mathbf{A}$ and $\mathbf{C}$ are similar (*see demos in the book*):

- The same determinant: $\det \mathbf{A} = \det \mathbf{C}$
- The same characteristic polynomial: $|\mathbf{A} - \lambda\mathbf{I}| = |\mathbf{C} - \lambda\mathbf{I}|$
- The same eigenvalues (with the same *algebraic* and *geometric* multiplicities).
- The same trace: $\left(\mathbf{A}\mathbf{I}_{\boldsymbol{\tau}_{[(\alpha)i+j]}}\right)_{|j} = \alpha(\mathbf{A}_{|i}) + \mathbf{A}_{|j} \quad \Rightarrow \quad {}_{j|}\left(\mathbf{A}\mathbf{I}_{\boldsymbol{\tau}_{[(\alpha)i+j]}}\right)_{|j} = \alpha({}_{j|}\mathbf{A}_{|i}) + {}_{j|}\mathbf{A}_{|j}$

***Mirror* inverse transf.:** $\left(\mathbf{I}_{(\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k)}\right)^{-1} = {}_{esp(\boldsymbol{\tau}_k^{-1}\cdots\boldsymbol{\tau}_1^{-1})}\mathbf{I}$

$$\mathbf{I} = {}_{\boldsymbol{\tau}_{[(-\alpha)j+i]}}\mathbf{I}_{\boldsymbol{\tau}_{[(\alpha)i+j]}} = {}_{\boldsymbol{\tau}_{[(\frac{1}{\alpha})j]}}\mathbf{I}_{\boldsymbol{\tau}_{[(\alpha)ij]}} \quad \Rightarrow \quad \mathbf{A} \text{ similar to } {}_{esp(\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k)^{-1}}\mathbf{A}_{\boldsymbol{\tau}_1\cdots\boldsymbol{\tau}_k}$$

F13

(**Lecture** 17) S-3 **Block diagonalization of a matrix (toothed matrix)**

Consider $\mathbf{A} = \left[\begin{array}{c||c} \mathbf{C} & \\ \hline \ast & \mathbf{L} \end{array}\right] \in \mathbb{C}^{n\times n}$ where

$\mathbf{C}$ (of order m) is *singular* and $\mathbf{L}$ is *full rank lower triangular*; then there exists an invertible $\mathbf{R}$ such that

$$\mathbf{R}^{-1}\mathbf{A}\mathbf{R} = \left[\begin{array}{c|c||l} & 0 & \\ \ast & \vdots & \\ {\scriptstyle m\times(m-1)} & 0 & \\ \hline & d_{m+1} & \beta_{m+1} \\ & d_{m+2} & \ast \quad \beta_{m+2} \\ \ast & \vdots & \ast \quad \ast \quad \ddots \\ & d_n & \ast \quad \ast \quad \cdots \quad \beta_n \end{array}\right]$$

$$\left(\ldots_{[(-\alpha_j)\overset{\tau}{m+j}]}\ldots\right)\mathbf{A}\left(\ldots_{[(\alpha_j)\overset{\tau}{j+m}]}\ldots\right); \qquad \boldsymbol{j} = 1,\ldots,m-1.$$

F14

(**Lecture** 17) S-4 **Block diagonalization of a matrix (toothed matrix)**

Consider $\mathbf{A} = \left[\begin{array}{c||c} \mathbf{C} & \\ \hline \ast & \mathbf{L} \end{array}\right] \in \mathbb{C}^{n\times n}$ where

$\mathbf{C}$ (of order m) is singular and $\mathbf{L}$ is full rank lower triangular, then there exists $\mathbf{S} = \mathbf{RP}$ (invertible) such that

$$\mathbf{P}^{-1}\mathbf{R}^{-1}\mathbf{A}\mathbf{R}\mathbf{P} = \left[\begin{array}{c|c||l} & 0 & \\ \ast & \vdots & \\ {\scriptstyle m\times(m-1)} & 0 & \\ \hline & 0 & \beta_{m+1} \\ & 0 & \ast \quad \beta_{m+2} \\ \ast & \vdots & \ast \quad \ast \quad \ddots \\ & 0 & \ast \quad \ast \quad \cdots \quad \beta_n \end{array}\right]$$

$$\left(\ldots_{[(-\alpha_j)\overset{\tau}{m+j}]}\ldots\right)\mathbf{R}^{-1}\mathbf{A}\mathbf{R}\left(\ldots_{[(\alpha_j)\overset{\tau}{j+m}]}\ldots\right); \qquad \boldsymbol{j} = m+1,\ldots,n.$$

F15

(**Lecture 17**) **S-5** **A very simple example**

Example 1. Consider $\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -2 & 1 \end{bmatrix}$ with eigenvalues 0, 1 and 1.

$$\left[\begin{array}{c}\mathbf{A}\\ \hline \mathbf{I}\end{array}\right] \xrightarrow[0\mathbf{I}]{(-)} \left[\begin{array}{ccc} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -2 & 1 \\ \hline 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[(1)\mathbf{1}+\mathbf{2}]\\ [(2)\mathbf{3}+\mathbf{2}]\\ [\mathbf{2}\rightleftharpoons\mathbf{3}]}]{\boldsymbol{\tau}} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right] \xrightarrow[\substack{[\mathbf{2}\rightleftharpoons\mathbf{3}]\\ [(-2)\mathbf{2}+\mathbf{3}]\\ [(-1)\mathbf{2}+\mathbf{1}]}]{\boldsymbol{\tau}} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ \hline 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right] \xrightarrow[0\mathbf{I}]{(+)} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ \hline 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right] = \left[\begin{array}{c}\mathbf{C}\\ \hline \mathbf{S}\end{array}\right]$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{\text{diagonal}}$$

F16

(**Lecture 17**) **S-6** **A not so simple example**

Example 2. Consider $\mathbf{A} = \begin{bmatrix} -2 & 0 & 3 \\ 3 & -2 & -9 \\ -1 & 2 & 6 \end{bmatrix}$ with eigenvalues *1*, *1* and *0*.

$$\xrightarrow[1\mathbf{I}]{(-)} \left[\begin{array}{ccc} -3 & 0 & 3 \\ 3 & -3 & -9 \\ -1 & 2 & 5 \\ \hline 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[[(-2)\mathbf{2}+\mathbf{3}]]{\boldsymbol{\tau}_{[(1)\mathbf{1}+\mathbf{3}]}} \left[\begin{array}{ccc} -3 & 0 & 0 \\ 3 & -3 & 0 \\ -1 & 2 & 0 \\ \hline 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[[(-1)\mathbf{3}+\mathbf{1}]]{\boldsymbol{\tau}_{[(2)\mathbf{3}+\mathbf{2}]}} \left[\begin{array}{ccc} -2 & -2 & 0 \\ 1 & 1 & 0 \\ -1 & 2 & 0 \\ \hline 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[1\mathbf{I}]{(+)} \left[\begin{array}{ccc} -1 & -2 & 0 \\ 1 & 2 & 0 \\ -1 & 2 & 1 \\ \hline 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[1\mathbf{I}]{(-)} \left[\begin{array}{ccc} -2 & -2 & 0 \\ 1 & 1 & 0 \\ -1 & 2 & 0 \\ \hline 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[[(-1)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \left[\begin{array}{ccc} -2 & 0 & 0 \\ 1 & 0 & 0 \\ -1 & 3 & 0 \\ \hline 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[[(1)\mathbf{2}+\mathbf{1}]]{\boldsymbol{\tau}} \left[\begin{array}{ccc} -1 & 0 & 0 \\ 1 & 0 & 0 \\ -1 & 3 & 0 \\ \hline 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[1\mathbf{I}]{(+)} \left[\begin{array}{ccc} 0 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 3 & 1 \\ \hline 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[0\mathbf{I}]{(-)} \left[\begin{array}{ccc} 0 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 3 & 1 \\ \hline 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[(-1)\mathbf{2}+\mathbf{1}]\\ [(4)\mathbf{3}+\mathbf{1}]}]{\boldsymbol{\tau}} \left[\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \\ \hline 6 & -1 & 1 \\ -9 & 1 & -2 \\ 4 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[(-4)\mathbf{1}+\mathbf{3}]\\ [(1)\mathbf{1}+\mathbf{2}]}]{\boldsymbol{\tau}} \left[\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \\ \hline 6 & -1 & 1 \\ -9 & 1 & -2 \\ 4 & 0 & 1 \end{array}\right] \xrightarrow[0\mathbf{I}]{(+)} \left[\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \\ \hline 6 & -1 & 1 \\ -9 & 1 & -2 \\ 4 & 0 & 1 \end{array}\right]$$

F17

(**Lecture** 17) **S-7** **Every matrix is similar to a toothed matrix**

For every $\mathbf{A}$ there exists $\mathbf{S}$ such that

$$\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \mathbf{C} \quad \Rightarrow \quad \mathbf{A}\mathbf{S} = \mathbf{S}\mathbf{C}$$

where $\mathbf{C}$, toothed, has the eigenvalues on the diagonal

Example 3.

$$\begin{bmatrix} 6 & -1 & 1 \\ -9 & 1 & -2 \\ 4 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} -2 & 0 & 3 \\ 3 & -2 & -9 \\ -1 & 2 & 6 \end{bmatrix} \begin{bmatrix} 6 & -1 & 1 \\ -9 & 1 & -2 \\ 4 & 0 & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}}_{\text{toothed}}$$

Consequences

- $\sum \lambda_i = \operatorname{tr}(\mathbf{A})$ and $\prod \lambda_i = \det \mathbf{A}$
- $\mathbf{A}\mathbf{S}_{|j} = \mathbf{S}\mathbf{C}_{|j} \quad \Rightarrow$ for j such that $\mathbf{C}_{|j} = \lambda_i \mathbf{I}_{|j}$:

$$\mathbf{A}(\mathbf{S}_{|j}) = \mathbf{S}(\lambda_i \mathbf{I}_{|j}) = \lambda_i(\mathbf{S}_{|j}) \quad \Rightarrow \quad \mathbf{S}_{|j} \text{ is an eigenvector.}$$

F18

(**Lecture** 17) **S-8** **Back to the simple, "toothless" example**

Consider $\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -2 & 1 \end{bmatrix}$ with eigenvalues 0, 1 and 1.

$$\left[\begin{array}{c} \mathbf{A} \\ \hline \mathbf{I} \end{array}\right] \xrightarrow{(-)0\mathbf{I}} \left[\begin{array}{ccc} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -2 & 1 \\ \hline 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[(1)\mathbf{1}+\mathbf{2}] \\ [(2)\mathbf{3}+\mathbf{2}] \\ [\mathbf{2}\rightleftharpoons\mathbf{3}]}]{\boldsymbol{\tau}} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right] \xrightarrow[\substack{[\mathbf{2}\rightleftharpoons\mathbf{3}] \\ [(-2)\mathbf{2}+\mathbf{3}] \\ [(-1)\mathbf{2}+\mathbf{1}]}]{\boldsymbol{\tau}} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ \hline 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right] \xrightarrow{(+)0\mathbf{I}} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ \hline 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right] = \left[\begin{array}{c} \mathbf{C} \\ \hline \mathbf{S} \end{array}\right]$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{A}(\mathbf{S}_{|j}) = \lambda_i(\mathbf{S}_{|j}) \quad \Rightarrow \quad \mathbf{S}_{|j} \text{ is an eigenvector.}$$

F19

(**Lecture** 17) **S-9** **Diagonalizable matrices**

- A matrix is diagonalizable **if and only if** *algebraic* and *geometric* multiplicities are equal for each eigenvalue
- If there are no repeated eigenvalues, there are no "teeth" either
- When there are no repeated eigenvalues $\underset{n\times n}{\mathbf{A}}$ is diagonalizable (is guaranteed to have n independent eigenvectors)

F20

(**Lecture 17**) **S-10** **Diagonalizing a matrix**

- Find the spectrum: $\{\lambda_1, \lambda_2, \ldots\}$
- Find the *algebraic multiplicity* of each eigenvalue: $\mu(\lambda_i)$

then choose one of these alternatives:

1. teething the matrix (implemented in NAcAL)
2. ...or for each λ_i
   - find the eigenspace
   $$\mathcal{E}_{\lambda_i}(\mathbf{A}) = \left\{ \boldsymbol{x} \in \mathbb{R}^n \,\middle|\, \mathbf{A}\boldsymbol{x} = \lambda_i \boldsymbol{x} \right\} \;=\; \mathcal{N}(\mathbf{A} - \lambda_i \mathbf{I}).$$
   - check $\mu(\lambda_i) = \dim \mathcal{E}_{\lambda_i}(\mathbf{A})$ (algebraic and geometric multiplicities are equal)

$$\mathbf{D} = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_k \end{bmatrix}; \quad \mathbf{S} = \Big[\text{Basis for } \mathcal{E}_{\lambda_1}(\mathbf{A}) \mathbin{+\!\!+} \cdots \mathbin{+\!\!+} \text{Basis for } \mathcal{E}_{\lambda_k}(\mathbf{A}) \Big]$$

$$\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \mathbf{D} \quad \Leftrightarrow \quad \mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$$

F21

(**Lecture 17**) **S-11** **Powers of a Diagonalizable Matrix**

If $\mathbf{A}\boldsymbol{x} = \lambda\boldsymbol{x}$ then $\mathbf{A}^2\boldsymbol{x} = \mathbf{A}\mathbf{A}\boldsymbol{x} = \mathbf{A}(\lambda\boldsymbol{x}) = \lambda\mathbf{A}\boldsymbol{x} =$

- What is the relationship between the eigenvectors of $\mathbf{A}$ and those of $\mathbf{A}^2$?
- What is the relationship between the eigenvalues of $\mathbf{A}$ and those of $\mathbf{A}^2$?

If $\mathbf{A}$ is diagonalizable and invertible, we have that

$$\mathbf{A}^{-1} = \left(\mathbf{S}\mathbf{D}(\mathbf{S}^{-1})\right)^{-1} = \mathbf{S}\mathbf{D}^{-1}(\mathbf{S}^{-1}).$$

Moreover, if we follow the following convention

$$\mathbf{A}^0 = \mathbf{A}^{(1-1)} = \mathbf{A}^1(\mathbf{A}^{-1}) = \mathbf{I};$$

then, for every diagonalizable and invertible $\mathbf{A}$ and every integer n: $\mathbf{A}^n = \mathbf{S}\mathbf{D}(\mathbf{S}^{-1}) \cdot \cdots \cdot \mathbf{S}\mathbf{D}(\mathbf{S}^{-1}) = \mathbf{S}\mathbf{D}^n(\mathbf{S}^{-1})$. F22

The lecture ends here

Questions of the Lecture 17

(L-17) Question 1. Factor these matrices into $\mathbf{S}\mathbf{D}\mathbf{S}^{-1}$;

(a) $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$

(b) $\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$

(Strang, 2006, exercise 15 from section 5.2.)

(L-17) Question 2. Which of these matrices cannot be diagonalized?

(a)
$$\mathbf{A}_1 = \begin{bmatrix} 2 & -2 \\ 2 & -2 \end{bmatrix}$$

(b)
$$\mathbf{A}_2 = \begin{bmatrix} 2 & 0 \\ 2 & -2 \end{bmatrix}$$

(c)
$$\mathbf{A}_3 = \begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix}$$

(Strang, 2006, exercise 5 from section 5.2.)

(L-17) Question 3. If $\mathbf{A} = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$ find $\mathbf{A}^{100}$ by diagonalizing $\mathbf{A}$.
(Strang, 2006, exercise 7 from section 5.2.)

(L-17) Question 4. If the eigenvalues of $\underset{3\times 3}{\mathbf{A}}$ are 1, 1, and 2, which of the following statements can we affirm to be true?
(a) $\mathbf{A}$ is invertible.
(b) $\mathbf{A}$ is diagonalizable.
(c) $\mathbf{A}$ is not diagonalizable.
(Strang, 2006, exercise 11 from section 5.2.)

(L-17) Question 5. Let $\mathbf{A}$ be the matrix
$$\mathbf{A} = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix}$$
(a) (1^{pts}) Determine if $\mathbf{A}$ is diagonalizable, and if so, find a diagonal matrix $\mathbf{D}$ and a matrix $\mathbf{S}$ such that $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$.
(b) (0.5^{pts}) Compute $\left(\mathbf{A}^6\right)\boldsymbol{v}$, where $\boldsymbol{v} = \begin{pmatrix} 0, & 0, & 0, & 1, \end{pmatrix}$.
(c) (0.5^{pts}) Using the values obtained in the first part, justify that $\mathbf{A}$ is invertible.
(d) (0.5^{pts}) What is the relationship between the eigenvalues of $\mathbf{A}$ and those of $\mathbf{A}^{-1}$?

(L-17) Question 6. If $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$; then $\mathbf{A}^3 = (\quad)(\quad)(\quad)$ and $\mathbf{A}^{-1} = (\quad)(\quad)(\quad)$.
(Strang, 2006, exercise 16 from section 5.2.)

(L-17) Question 7. Consider the matrix
$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$$
(a) Find the eigenvalues of $\mathbf{A}$.
(b) Find the eigenvectors of $\mathbf{A}$.
(c) Diagonalize $\mathbf{A}$: write it as $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$.

(L-17) Question 8. False or true? If the eigenvalues of $\mathbf{A}$ are 2, 2, and 3, then we know that the matrix is
(a) Invertible
(b) Diagonalizable
(c) Not diagonalizable.

(L-17) Question 9. Let the matrices
$$\mathbf{A}_1 = \begin{bmatrix} 8 & \\ & 2 \end{bmatrix}; \qquad \mathbf{A}_2 = \begin{bmatrix} 9 & 4 \\ & 1 \end{bmatrix}; \qquad \mathbf{A}_3 = \begin{bmatrix} 10 & 5 \\ -5 & \end{bmatrix}$$

(a) Complete these matrices so that in all three cases $\det \mathbf{A}_i = 25$. Thus, the trace is equal to 10 in all cases, and therefore for the three matrices, the only eigenvalue $\lambda = 5$ is repeated twice ($\lambda^2 = 25$ and $\lambda + \lambda = 10$ imply $\lambda = 5$).
(b) Find a characteristic vector with $\mathbf{A}\boldsymbol{x} = 5\boldsymbol{x}$. These three matrices are not diagonalizable because there is not a second linearly independent eigenvector from the first.

(Strang, 2006, exercise 27 from section 5.2.)

(L-17) Question 10. Factor the following matrices into $\mathbf{S}\,\mathbf{D}\,\mathbf{S}^{-1}$.

(a) $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

(b) $\mathbf{B} = \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$

(Strang, 2006, exercise 1 from section 5.2.)

(L-17) Question 11. Find the matrix $\mathbf{A}$ whose eigenvalues are 1 and 4, whose eigenvectors are $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ respectively.
(Strang, 2006, exercise 2 from section 5.2.)

(L-17) Question 12. If the diagonal elements of an upper triangular matrix of order 3×3 are 1, 2, and 7, can you determine if the matrix is diagonalizable? What is $\mathbf{D}$?
(Strang, 2006, exercise 4 from section 5.2.)

(L-17) Question 13.

(a) Find the eigenvalues and eigenvectors of the matrix $\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$.
(b) Explain why (or why not) the matrix $\mathbf{A}$ is diagonalizable.

(L-17) Question 14. Let $\mathbf{A}$ be a 3×3 matrix. Assume that its eigenvalues are 1 and 0, that a basis of the eigenvectors associated with $\lambda = 1$ are $[1, 0, 1]$ and $[0, 0, 1]$; while those associated with $\lambda = 0$ are parallel to $[1, 1, 2]$.
(a) Is $\mathbf{A}$ diagonalizable? If so, write the diagonal matrix $\mathbf{D}$ and the matrix $\mathbf{S}$ such that $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$.
(b) Find $\mathbf{A}$.

(L-17) Question 15. Let $\mathbf{A}$ be a 2×2 matrix such that $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ is an eigenvector for $\mathbf{A}$ with eigenvalue 2, and $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ is another eigenvector for $\mathbf{A}$ with eigenvalue -2. If $\boldsymbol{v} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, compute $\left(\mathbf{A}^3\right)\boldsymbol{v}$.

End of Questions of the Lecture 17

LECTURE 18: Symmetric Matrices and Orthogonal Diagonalization

Lecture 18

(**Lecture** 18) S-1 Highlights of Lesson 18

Highlights of *Lesson 18*

- Symmetric matrices $\mathbf{A} = \mathbf{A}^{\mathsf{T}}$
  - Eigenvalues and eigenvectors
- Introd. positive definite matrices

F23

(**Lecture** 18) S-2 Symmetric Matrices $\mathbf{A} = \mathbf{A}^{\mathsf{T}}$

What is special about $\mathbf{A}\boldsymbol{x} = \lambda\boldsymbol{x}$ when $\underset{n\times n}{\mathbf{A}}$ is symmetric?

1. The eigenvalues are REAL
2. n eigenvectors *can be chosen* ORTHOGONAL

(Always diagonalizable)

The usual diagonalizable case:

$$\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \mathbf{D} \quad \longleftrightarrow \quad \mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$$

Symmetric case:
I can choose orthonormal eigenvectors (ortho*normal* columns of $\mathbf{S} = \mathbf{Q}$)

$$(\text{If } \mathbf{A} = \mathbf{A}^{\mathsf{T}}) \qquad \mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{-1} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\mathsf{T}} \qquad \text{Spectral theorem}$$

Orthogonally diagonalizable.

F24

(**Lecture** 18) S-3 Eigenspaces are orthogonal for symmetric matrices

Eigenvectors (corresponding to distinct eigenvalues) of a symmetric matrix are orthogonal.

Proof. Consider $\mathbf{A}\boldsymbol{x} = \lambda_1\boldsymbol{x}$ and $\mathbf{A}\boldsymbol{y} = \lambda_2\boldsymbol{y}$ (with $\lambda_1 \neq \lambda_2$). Then

$$\lambda_1\boldsymbol{x}\cdot\boldsymbol{y} \;=\; \mathbf{A}\boldsymbol{x}\cdot\boldsymbol{y} \;=\; \boldsymbol{x}(\mathbf{A}^{\mathsf{T}})\boldsymbol{y} \;=\; \boldsymbol{x}\mathbf{A}\boldsymbol{y} \;=\; (\boldsymbol{x}\cdot\boldsymbol{y})\lambda_2.$$

Since $\lambda_1 \neq \lambda_2$ necessarily:

$$\lambda_1(\boldsymbol{x}\cdot\boldsymbol{y}) - \lambda_2(\boldsymbol{x}\cdot\boldsymbol{y}) = 0 \quad \Longrightarrow \quad (\lambda_1 - \lambda_2)\boldsymbol{x}\cdot\boldsymbol{y} = 0 \quad \Longrightarrow \quad \boldsymbol{x}\cdot\boldsymbol{y} = 0.$$

□

F25

(Lecture 18) S-4 Quadratic forms

Quadratic form:

$$\boldsymbol{x}\mathbf{A}\boldsymbol{x}; \quad \text{with} \quad \mathbf{A}^\intercal = \mathbf{A}$$

Since $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\intercal$ (with $\mathbf{Q}^\intercal\mathbf{Q} = \mathbf{Q}\mathbf{Q}^\intercal = \mathbf{I}$), then

$$\boldsymbol{x}\mathbf{A}\boldsymbol{x} = \boldsymbol{x}\mathbf{Q}\mathbf{D}\mathbf{Q}^\intercal\boldsymbol{x} = (\mathbf{Q}^\intercal\boldsymbol{x})\mathbf{D}(\mathbf{Q}^\intercal\boldsymbol{x}) \text{ (weighted sum of squares)}$$

Positive definite quadratic form:

$$\boldsymbol{x}\mathbf{A}\boldsymbol{x} > 0 \quad \forall \boldsymbol{x} \neq \mathbf{0} \qquad \Longleftrightarrow \qquad \lambda_i > 0, \quad i = 1:n.$$

Then we also say that $\mathbf{A}$ is positive definite.

F26

(Lecture 18) S-5 Positive definite matrices

Meaning:

$$\boldsymbol{x}\mathbf{A}\boldsymbol{x} > 0 \qquad \text{(except for } \boldsymbol{x} = \mathbf{0})$$

Some properties

Consider a positive definite matrix $\mathbf{A}$: What about $\mathbf{A}^{-1}$?

$\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{-1} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\intercal$

Consider two positive definite matrices $\mathbf{A}$ and $\mathbf{B}$: What about $\mathbf{A} + \mathbf{B}$?

Thus the answer is...

F27

(Lecture 18) S-6 Matrix Product $\mathbf{A}^\intercal\mathbf{A}$

Let $\underset{m\times n}{\mathbf{A}}$ be a rectangular matrix. Is $\mathbf{A}^\intercal\mathbf{A}$ positive definite?

$$\boldsymbol{x}(\mathbf{A}^\intercal\mathbf{A})\boldsymbol{x} =$$

It can only be 0 when $\mathbf{A}\boldsymbol{x}$ is $\mathbf{0}$
How can we guarantee that $\mathbf{A}\boldsymbol{x} \neq \mathbf{0}$ when $\boldsymbol{x} \neq \mathbf{0}$?

F28

(Lecture 18) S-7 Symmetric matrices: Signs of eigenvalues

Are all λ_i positive? Are they negative?
Computing eigenvalues of $\underset{5\times 5}{\mathbf{A}}$ is impossible in general! (5th degree polynomial)

Good news: The signs of the pivots of echelon form are the same as the signs of the eigenvalues λ_i
(If we do not change the sign of the determinant with *Type II* elementary transformations)

Number of positive pivots = Number of positive eigenvalues

F29

(**Lecture** 18) S-8 **Positive definite symmetric matrices**

- All eigenvalues are:
- All pivots are:

$$\begin{bmatrix} 5 & 2 \\ 2 & 3 \end{bmatrix}$$

Pivots:

What is the sign of each eigenvalue?

$$\lambda^2 - 8\lambda + 11 = 0 \rightarrow \qquad \lambda = 4 \pm \sqrt{5} > 0$$

F30

Summary (for symmetric matrices):

1. Symmetric matrices have *real eigenvalues* and *perpendicular eigenvectors* can be chosen
2. $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\intercal$ where $\mathbf{Q}$ is orthogonal
3. $\mathbf{A}$ is symmetric if and only if it is *orthogonally* diagonalizable
4. The signs of the pivots in the echelon form are the same as the signs of the eigenvalues λ_i (only if we do not change the sign of the determinant with *Type II* elementary transformations)

The lecture ends here

Questions of the Lecture 18

(L-18) Question 1. Write the matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ in the form $\mathbf{Q}\mathbf{D}\mathbf{Q}^\intercal$ of the spectral theorem:

(a) $\mathbf{A} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$

(b) $\mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(c) $\mathbf{C} = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$

(Strang, 2006, exercise 11 from section 5.5.)

(L-18) Question 2. Find the eigenvalues and the unit eigenvectors (of unit length) of

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

(Strang, 2003, exercise 3 from section 6.4.)

(L-18) Question 3. Find an orthonormal matrix $\mathbf{Q}$ that diagonalizes this symmetric matrix:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & -2 \\ 2 & -2 & 0 \end{bmatrix}$$

(Strang, 2003, exercise 5 from section 6.4.)

(L-18) Question 4. Suppose that $\mathbf{A}$ is a symmetric 3×3 matrix with eigenvalues 0, 1, and 2.

(a) What properties can be guaranteed for the unit eigenvectors $\boldsymbol{u}$, $\boldsymbol{v}$, and $\boldsymbol{w}$ corresponding to the respective eigenvalues 0, 1, and 2?
(b) In terms of $\boldsymbol{u}$, $\boldsymbol{v}$, and $\boldsymbol{w}$, describe the null space $\mathcal{N}\left(\mathbf{A}\right)$, the left null space $\mathcal{N}\left(\mathbf{A}^{\intercal}\right)$, the row space $\mathcal{C}\left(\mathbf{A}^{\intercal}\right)$, and the column space $\mathcal{C}\left(\mathbf{A}\right)$.
(c) Find a vector $\boldsymbol{x}$ such that $\mathbf{A}\boldsymbol{x} = \boldsymbol{v} + \boldsymbol{w}$. Is $\boldsymbol{x}$ unique?
(d) What conditions must we impose on $\boldsymbol{b}$ for $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ to have a solution?
(e) If $\boldsymbol{u}$, $\boldsymbol{v}$, and $\boldsymbol{w}$ are the columns of $\mathbf{S}$, and $\boldsymbol{v}$ is orthogonal to $\boldsymbol{w}$; write the matrices $\mathbf{S}^{-1}$ and $\mathbf{S}^{-1}\mathbf{A}\mathbf{S}$.

(Strang, 2006, exercise 13 from section 5.5.)

(L-18) Question 5. Write a key fact about the eigenvalues of each of these types of matrices:

(a) A real symmetric matrix.
(b) A diagonalizable matrix such that $\mathbf{A}^n \to \mathbf{0}$ as $n \to \infty$.
(c) A non-diagonalizable matrix.
(d) A singular matrix.

(Strang, 2006, exercise 16 from section 5.5.)

(L-18) Question 6. Let

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}; \qquad \mathbf{B} = \frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

(a) Find the eigenvalues of $\mathbf{A}$ (remember that $i^2 = -1$).
(b) Find the eigenvalues of $\mathbf{B}$ (in this case, it may be easier to first find the eigenvectors, and then deduce the eigenvalues).

(Strang, 2006, exercise 14 from section 5.5.)

(L-18) Question 7. If $\mathbf{A}^3 = \mathbf{0}$ then the eigenvalues of $\mathbf{A}$ must be _____. Provide an example such that $\mathbf{A} \neq \mathbf{0}$. Furthermore, if $\mathbf{A}$ is also symmetric, prove that $\mathbf{A}^3$ is necessarily $\mathbf{0}$.

(L-18) Question 8. Consider the matrix

$$\mathbf{A} = \begin{bmatrix} a & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}.$$

(a) Prove that $\mathbf{A}$ is not diagonalizable when $a = 3$.
(b) Is $\mathbf{A}$ diagonalizable when $a = 2$? (explain). If it is diagonalizable, find an eigenvalue diagonal matrix $\mathbf{D}$ and an eigenvector matrix $\mathbf{S}$ such that $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$.
(c) Is $\mathbf{A}^{\intercal}\mathbf{A}$ diagonalizable for any value of a? Is it possible to find a full set of orthonormal eigenvectors of $\mathbf{A}^{\intercal}\mathbf{A}$?
(d) Find all values of a for which $\mathbf{A}$ is invertible and diagonalizable.

(L-18) Question 9. Sea la matriz

$$\mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix};$$

(a) Exprese $\mathbf{B}$ en la forma $\mathbf{B} = \mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\intercal}$ del teorema espectral.
(b) ¿Es $\mathbf{B}$ diagonalizable? Si no lo es, diga las razones; y en caso contrario genere una matriz $\mathbf{S}$ que diagonalice a $\mathbf{B}$.

(L-18) Question 10. Suppose the vectors $\boldsymbol{q}_1$, $\boldsymbol{q}_2$, $\boldsymbol{q}_3$ form an orthonormal basis for $\mathbb{R}^3$ and the matrix $\mathbf{A}$ satisfies $\mathbf{A}\boldsymbol{q}_1 = (1, 0, 0,)$, $\mathbf{A}\boldsymbol{q}_2 = (0, 1, 0,)$, and $\mathbf{A}\boldsymbol{q}_3 = (0, 0, 1,)$.

(a) (0.5^{pts}) Write the matrix $\mathbf{A}$ explicitly in terms of the vectors $\boldsymbol{q}_1$, $\boldsymbol{q}_2$, $\boldsymbol{q}_3$.
(b) (1^{pts}) Write down all possibilities for $\det \mathbf{A}$.
(c) Which of the following statements are correct: The eigenvalues of $\mathbf{A}$ must...

- be real numbers.
- be positive real numbers.
- be imaginary numbers.
- have absolute value $|\lambda| = 1$.

End of Questions of the Lecture 18

LECTURE 19: Diagonalization by congruence. Definite Matrices

Lecture 19

(**Lecture 19**) S-1 Highlights of Lesson 19

Highlights of *Lesson 19*

- Positive and Negative (semi)definite matrices
- Completing the square
- Diagonalization by congruence

F32

(**Lecture 19**) S-2 Quadratic forms

- Positive definite: $\forall \boldsymbol{x} \neq \mathbf{0} \;\Rightarrow\; \boldsymbol{x}\mathbf{A}\boldsymbol{x} > 0$.
- Positive semi-definite: $\forall \boldsymbol{x} \neq \mathbf{0} \;\Rightarrow\; \boldsymbol{x}\mathbf{A}\boldsymbol{x} \geq 0$.
- Negative definite: $\forall \boldsymbol{x} \neq \mathbf{0} \;\Rightarrow\; \boldsymbol{x}\mathbf{A}\boldsymbol{x} < 0$.
- Negative semi-definite: $\forall \boldsymbol{x} \neq \mathbf{0} \;\Rightarrow\; \boldsymbol{x}\mathbf{A}\boldsymbol{x} \leq 0$.
- Indefinite: neither positive semi-definite nor negative semi-definite.

F33

Example 4. What number do I have to put for the matrix $\mathbf{A}$ to be singular?

$$\mathbf{A} = \begin{bmatrix} 2 & 6 \\ 6 & \end{bmatrix}$$

- Eigenvalues:
- Leading principal minors:
- For the quadratic form

$$q_{\mathbf{A}}(\boldsymbol{x}) = \boldsymbol{x}\mathbf{A}\boldsymbol{x} = \begin{pmatrix} x, & y, \end{pmatrix} \begin{bmatrix} 2 & 6 \\ 6 & \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2x^2 + 12xy + \quad y^2$$

Is there a $\boldsymbol{x} \neq \mathbf{0}$ such that $\boldsymbol{x}\mathbf{A}\boldsymbol{x} = 0$?

Example 5.

$$\text{If } \mathbf{A} = \begin{bmatrix} 2 & 6 \\ 6 & 7 \end{bmatrix} \quad \text{then} \quad \begin{pmatrix} x, & y, \end{pmatrix} \begin{bmatrix} 2 & 6 \\ 6 & 7 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2x^2 + 12xy + \quad y^2$$

- Are there numbers x and y that make $\boldsymbol{x}\mathbf{A}\boldsymbol{x}$ negative?
- Does the function go through the origin?
- When $y = 0$ and $x = 1$, is it positive? (and when $x = -1$?)

- When $x = 0$ and $y = 1$, is it positive? (and when $y = -1$?)
- Is it always positive?

$(0, 0,)$ **saddle point**: minimum in some directions, maximum in others.

$$\lambda_1 = -2, \quad \begin{pmatrix} -6 \\ 4 \end{pmatrix}; \qquad \lambda_2 = 11, \quad \begin{pmatrix} 6 \\ 9 \end{pmatrix}$$

Example 6.

$$\text{If } \mathbf{A} = \begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix} \text{ then } \begin{pmatrix} x, & y, \end{pmatrix} \begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2x^2 + 12xy + 20y^2$$

Positive definite.

Tests to check if $\mathbf{A}$ is positive definite

- Are the leading principal minors positive?
- Are the eigenvalues positive?

$$q_{\mathbf{A}}(\boldsymbol{x}) = \boldsymbol{x}\mathbf{A}\boldsymbol{x} > 0 \qquad \text{for all } \boldsymbol{x} \neq \mathbf{0}$$

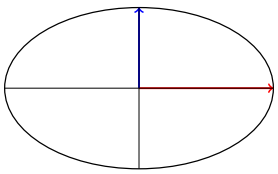
(**Lecture 19**) **S-3 Completing the squares**

If we could express $q(\boldsymbol{x})$ as a sum of squares, we would know whether $q(\boldsymbol{x})$ is positive definite.

Let's complete the square!

- $q(x, y) = 2x^2 + 12xy + 20y^2 \;=\; 2(x + \quad y)^2 +$
- $q(x, y) = 2x^2 + 12xy + 7y^2$
- $q(x, y) = 2x^2 + 12xy + 18y^2$
- $q(x, y) = 2x^2 + 12xy + 200y^2$ (graph)

If it is positive definite: $q(x, y) = a$; $\quad a > 0$: ellipse



is $q(x, y, z, w, t) = 2t^2 - 2tx - 2tz + w^2 - 2wy + 2x^2 - 2xy + 2y^2 + z^2$ positive definite? ☹☹☹☹☹!!!???

F37

```
x, y, z, w, t  = sympy.symbols('x y z w t')
B = Matrix([[-1,1,0,0,0],[1,0,0,0,-1],[0,-1,0,1,0],[0,0,1,0,-1]])
A = ~B*B
v = Vector([x,y,z,w,t])
q = sympy.factor(v*A*v)
D = DiagonalizaC(A)
dispElimFyC(A,D.pasos)
q
```

(**Lecture** 19) S-4 **Congruent matrices**

$\mathbf{A}$ and $\mathbf{C}$ are congruent if there exists an invertible $\mathbf{B}$ such that $\boxed{\mathbf{C} = \mathbf{B}^\intercal\mathbf{A}\mathbf{B}}$

Diagonalization by congruence
For every $\mathbf{A}$ (symmetric) there exists $\mathbf{B} = \mathbf{I}_{\tau_1\cdots\tau_k}$ *(invertible)* such that

$$\mathbf{D} = \mathbf{B}^\intercal\mathbf{A}\mathbf{B} \qquad \text{is diagonal} \qquad (\mathbf{B}^\intercal = {}_{\tau_k\cdots\tau_1}\mathbf{I})$$

Spectral Theorem: ¡Diagonalization by similarity and congruence!

$$\mathbf{D} = \mathbf{Q}^{-1}\mathbf{A}\mathbf{Q} = \mathbf{Q}^\intercal\mathbf{A}\mathbf{Q}.$$

Hence, every quadratic form can be expressed as a sum of squares.

$$\boldsymbol{x}\mathbf{A}\boldsymbol{x} \;=\; \boldsymbol{x}\left(\mathbf{B}^{-1}\right)^\intercal\mathbf{D}\mathbf{B}^{-1}\boldsymbol{x} \;=\; \boldsymbol{y}\mathbf{D}\boldsymbol{y}; \qquad \text{where} \qquad \boldsymbol{y} = \mathbf{B}^{-1}\boldsymbol{x}.$$

F38

(**Lecture** 19) S-5 **Completing the squares**

$$2x^2 + 12xy + 20y^2$$

$$\begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix} \xrightarrow[[(-3)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \begin{bmatrix} 2 & 0 \\ 6 & 2 \end{bmatrix} \xrightarrow[[(-3)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix};$$

therefore, we have that:

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \mathbf{D} = \mathbf{E}^\intercal\mathbf{A}\mathbf{E} = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}\begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix}\begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}$$

hence $\mathbf{A} = (\mathbf{E}^\intercal)^{-1}\mathbf{D}\mathbf{E}^{-1}$ and so

$$\begin{aligned} \boldsymbol{x}\mathbf{A}\boldsymbol{x} &= (x, \quad y,)\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \left(\boldsymbol{x}\left(\mathbf{E}^{-1}\right)^\intercal\right)\mathbf{D}\left(\mathbf{E}^{-1}\boldsymbol{x}\right) \\ &= \left((x+3y), \quad y,\right)\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}\begin{pmatrix} (x+3y) \\ y \end{pmatrix} = 2(x+3y)^2 + 2y^2 \end{aligned}$$

F39

(**Lecture** 19) S-6 **Example 3 by 3**

$$\text{Is } \mathbf{A} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \text{ positive definite?}$$

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \xrightarrow[\tau\,[(\frac{1}{2})\mathbf{1+2}]]{\tau\,[(\frac{1}{2})\mathbf{1+2}]} \begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{3}{2} & -1 \\ 0 & -1 & 2 \end{bmatrix} \xrightarrow[\tau\,[(\frac{2}{3})\mathbf{2+3}]]{\tau\,[(\frac{2}{3})\mathbf{2+3}]} \begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{3}{2} & 0 \\ 0 & 0 & \frac{4}{3} \end{bmatrix}$$

$$\boldsymbol{x}\mathbf{A}\boldsymbol{x} = 2x^2 + 2y^2 + 2z^2 - 2xy - 2yz > 0$$

$\boldsymbol{x}\mathbf{A}\boldsymbol{x} = 1$: (ellipsoid) Axes are in the direction of the eigenvectors $\mathbf{A} = \mathbf{Q}^\intercal\boldsymbol{\lambda}\mathbf{Q}$ F40

If $\mathbf{B} = \mathbf{I}_{\tau\,[(\frac{1}{2})\mathbf{1+2}][(\frac{2}{3})\mathbf{2+3}]}$, that is, if

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow[\tau\,[(\frac{2}{3})\mathbf{2+3}]]{\tau\,[(\frac{1}{2})\mathbf{1+2}]} \begin{bmatrix} 1 & -1/2 & 1/3 \\ 0 & 1 & -2/3 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{B} \quad \Rightarrow \quad \mathbf{B}^{-1} = \begin{bmatrix} 1 & -1/2 & 0 \\ 0 & 1 & -2/3 \\ 0 & 0 & 1 \end{bmatrix},$$

we have that the quadratic form $\boldsymbol{x}\mathbf{A}\boldsymbol{x}$ can be expressed as:

$$\boldsymbol{x}\mathbf{A}\boldsymbol{x} = \begin{pmatrix} x, & y, & z, \end{pmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ 0 & -2/3 & 1 \end{bmatrix} \begin{bmatrix} 2 & & \\ & 3/2 & \\ & & 4/3 \end{bmatrix} \begin{bmatrix} 1 & -1/2 & 0 \\ 0 & 1 & -2/3 \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \boldsymbol{x}\left(\mathbf{B}^{-1}\right)^\intercal \mathbf{D}\mathbf{B}^{-1}\boldsymbol{x}$$

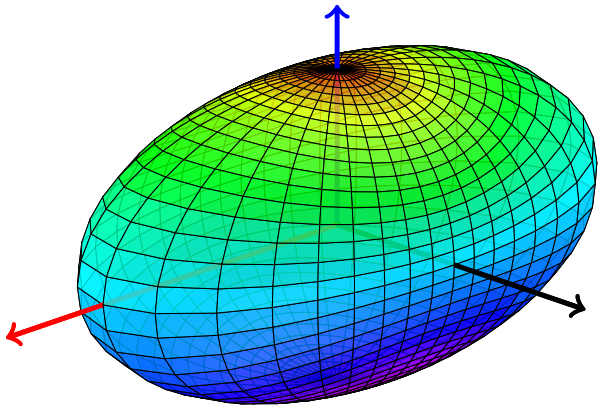
$$= \begin{pmatrix} (x - 1/2y), & (y - 2/3z), & z, \end{pmatrix} \begin{bmatrix} 2 & & \\ & 3/2 & \\ & & 4/3 \end{bmatrix} \begin{pmatrix} (x - 1/2y) \\ (y - 2/3z) \\ z \end{pmatrix} = \left(\mathbf{B}^{-1}\boldsymbol{x}\right) \cdot \mathbf{D} \cdot \left(\mathbf{B}^{-1}\boldsymbol{x}\right)$$

$$= 2\,(x - 1/2y)^2 \; + \; 3/2\,(y - 2/3z)^2 \; + \; 4/3z^2 \qquad = \sum\nolimits_j \mathbf{D}_{|j} \cdot \left(y_j\right)^2 \quad (\text{con } \; \boldsymbol{y} = \mathbf{B}^{-1}\boldsymbol{x});$$

is a sum of three squared terms, each multiplied by one of the pivots of $\mathbf{D}$; since all the pivots are positive, the quadratic form is evidently positive definite.

(**Lecture** 19) S-7 **Positive definite matrices and ellipsoids: example 3 by 3**

- The region $(\boldsymbol{x}\mathbf{A}\boldsymbol{x} = a)$ is an (ellipsoid).
- The columns of $\mathbf{Q}$ (eigenvectors of $\mathbf{A}$) are the principal axes.
- Lengths of the axes are determined by the eigenvalues.



F41

(**Lecture** 19) S-8 **Another example 3 by 3**

$$\text{Is } \mathbf{A} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \text{ positive definite?}$$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \xrightarrow[\boldsymbol{\tau}[(1)\mathbf{3}+\mathbf{1}]]{} \begin{bmatrix} 2 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \xrightarrow[\boldsymbol{\tau}[(-\frac{1}{2})\mathbf{1}+\mathbf{3}]]{} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} \end{bmatrix} \xrightarrow[\boldsymbol{\tau}[\mathbf{2}\rightleftharpoons\mathbf{3}]]{} \begin{bmatrix} 2 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Indefinite matrix.

F42

(**Lecture** 19) S-9 "Classification" of quadratic forms

$$\boldsymbol{x}\mathbf{A}\boldsymbol{x} \lessgtr 0; \quad \text{for all } \boldsymbol{x} \neq \mathbf{0}$$

Methods
Check the signs of

1. Elem. diag.: $\mathbf{D} = \mathbf{B}^\intercal\mathbf{A}\mathbf{B}$ (*Diagonalization by congruence*) ☺
2. Computing eigenvalues: (Roots of a polynomial) ☹
3. Leading principal minors: (Sylvester's criterion) ☹

Law of inertia
the number of positive, negative, and zero entries on the diagonal of $\mathbf{D}$ is an invariant of $\mathbf{A}$, i.e., it does not depend on $\mathbf{B}$

(Orthogonal diagonalization $\mathbf{D} = \mathbf{Q}^\intercal\mathbf{A}\mathbf{Q}$ is a special case) F43

Sylvester's criterion for minors indicates that if all principal minors are positive, then the matrix is positive definite. However, this criterion does not provide clues about the sign of the eigenvalues when any of the minors is zero. This is the case in the last example, where one eigenvalue is zero (with multiplicity 1); since the trace is zero but the rank is non-zero, the other two eigenvalues must be non-zero and of opposite signs. In other words, the matrix is indefinite, yet all three principal minors are zero!

There is a verification using minors for positive semidefinite matrices, but it is more complicated (and if the matrix is negative, there is an additional complication with the alternation of the signs of the minors). Therefore, it is much more practical to complete the square through congruence diagonalization: $\mathbf{D} = \mathbf{B}^\intercal\mathbf{A}\mathbf{B}$.

The lecture ends here

Questions of the Lecture 19

(L-19) Question 1. Decide whether the following matrices are positive definite, and write out the corresponding quadratic forms $f = \boldsymbol{x}\mathbf{A}\boldsymbol{x}$:

(a) $\begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$

(d) $\begin{bmatrix} -1 & 2 \\ 2 & -8 \end{bmatrix}$

(e) The determinant in (b) is zero; along what line does $f(x, y) = 0$?

(Strang, 2006, exercise 2 from section 6.1.)

(L-19) Question 2. What is the quadratic form $f = ax^2 + 2bxy + cy^2$ for each of these matrices? Complete the square to write f as a sum of one or two squares $d_1(\quad)^2 + d_2(\quad)^2$.

(a) $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 9 \end{bmatrix}$

(b) $\mathbf{B} = \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}$

(Strang, 2006, exercise 15 from section 6.1.)

(L-19) Question 3. Which of the following matrices has two positive eigenvalues? Test $a > 0$ and $ac > b^2$, do not compute the eigenvalues. $\boldsymbol{x}\mathbf{A}\boldsymbol{x} < 0$.

(a) $\mathbf{A} = \begin{bmatrix} 5 & 6 \\ 6 & 7 \end{bmatrix}$
(b) $\mathbf{B} = \begin{bmatrix} -1 & -2 \\ -2 & -5 \end{bmatrix}$
(c) $\mathbf{C} = \begin{bmatrix} 1 & 10 \\ 10 & 100 \end{bmatrix}$
(d) $\mathbf{D} = \begin{bmatrix} 1 & 10 \\ 10 & 101 \end{bmatrix}$
(Strang, 2006, exercise 14 from section 6.1.)

(L-19) Question 4. Show that $f(x, y) = x^2+4xy+3y^2$ does not have a minimum at $(0, 0)$ even though it has positive coefficients. Write $f(x, y)$ as a difference of squares and find a point (x, y) where $f(x, y)$ is negative.
(Strang, 2006, exercise 16 from section 6.1.)

(L-19) Question 5. Show from the eigenvalues that if $\mathbf{A}$ is positive definite, so is $\mathbf{A}^2$ and so is$\mathbf{A}^{-1}$.
(Strang, 2006, exercise 4 from section 6.2.)

(L-19) Question 6. Consider the following quadratic forms

$$\begin{aligned} q_1(x, y, z) &= x^2 + 4y^2 + 5z^2 - 4xy. \\ q_2(x, y, z) &= -x^2 + 4y^2 + z^2 + 2xy - 2axz. \end{aligned}$$

(a) Show that $q_1(x, y, z)$ is positive semi-definite.
(b) Find, if it is possible, any value of a such that $q_2(x, y, z)$ is negative definite.

(L-19) Question 7. Decide whether the following matrices are positive definite or not.
(a) $\mathbf{A} = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$
(b) $\mathbf{B} = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}$
(c) $\mathbf{C} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}^2$
(Strang, 2006, exercise 2 from section 6.2.)

(L-19) Question 8. Classify the quadratic form (i.e., positive definite, negative, semi-definite, indefinite, etc.)

$$q(x, y, z) = x^2 + 6xy + y^2 + az^2;$$

in terms of the parameter a.

(L-19) Question 9. If $\mathbf{A} = \left[\begin{smallmatrix} a & b \\ b & d \end{smallmatrix}\right]$ is positive definite, prove that $\mathbf{A}^{-1}$ is positive definite.
(Strang, 2006, exercise 8 from section 6.1.)

(L-19) Question 10. If a symmetric matrix of 2 by 2 satisfies $a > 0$, and $ac > b^2$, prove that its eigenvalues are real and positive (positive definite). Use the characteristic equation and the fact that the product of the eigenvalues equals the determinant.
(Strang, 2006, exercise 3 from section 6.1.)

(L-19) Question 11. Decide whether the following matrices are positive definite, negative definite, semi-definite, or indefinite.
(a) $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 4 \\ 3 & 4 & 9 \end{bmatrix}$

(b) $\mathbf{B} = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 2 & 6 & -2 & 0 \\ 0 & -2 & 5 & -2 \\ 0 & 0 & -2 & 3 \end{bmatrix}$
(c) $\mathbf{C} = -\mathbf{B}$
(d) $\mathbf{D} = \mathbf{A}^{-1}$

(L-19) Question 12. A positively defined matrix cannot have a zero (or worse; a negative number) on its main diagonal. Prove that this matrix does not satisfy. $\boldsymbol{x}\mathbf{A}\boldsymbol{x} > 0$, for all $\boldsymbol{x} \neq \mathbf{0}$:

$$\begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} \begin{bmatrix} 4 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 5 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \quad \textit{isnotpositivewhen} \quad \begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} = \begin{pmatrix} \quad & \quad & \quad \end{pmatrix}$$

(Strang, 2006, exercise 21 from section 6.2.)

(L-19) Question 13. Demonstrate that if $\mathbf{A}$ and $\mathbf{B}$ are positive definite then $\mathbf{A}+\mathbf{B}$ is also positive definite. For this demonstration, pivots and eigenvalues are not convenient. It is better to use $\boldsymbol{x}(\mathbf{A}+\mathbf{B})\boldsymbol{x} > 0$
(Strang, 2006, exercise 5 from section 6.2.)

(L-19) Question 14. Find the $\dot{\mathbf{L}}\mathbf{D}\dot{\mathbf{L}}^{\intercal}$ factorization for the following symmetric matrices.
(a)

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{bmatrix}$$

(L-19) Question 15. The quadratic form $f(x,y) = 3(x+2y)^2 + 4y^2$ is positive definite. Find the matrix $\mathbf{A}$, factor it into $\mathbf{L}\mathbf{D}\mathbf{L}^{\intercal}$, and relate the elements in $\mathbf{D}$ and $\mathbf{L}$ with 3, 2 and 4 in f.
(Strang, 2006, exercise 9 from section 6.1.)

(L-19) Question 16. Consider the following matrices

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & a & 0 \\ 0 & 0 & a & a \end{bmatrix} \qquad \text{and} \qquad \mathbf{B} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(a) (0.5^{pts}) Compute the eigenvalues of $\mathbf{A}$.
(b) (0.5^{pts}) Prove that when $a = 2$ the matrix $\mathbf{A}$ is not diagonalizable.
(c) (1^{pts}) For matrix $\mathbf{B}$, find a diagonal matrix $\mathbf{D}$ and an orthonormal matrix $\mathbf{P}$ such that $\mathbf{B} = \mathbf{P}\mathbf{D}\mathbf{P}^{\intercal}$.
(d) (0.5^{pts}) Find the polynomial expression of the quadratic form associated with the matrix $\mathbf{B}$, and prove that it is positive definite.

Versión de un ejercicio proporcionado por Mercedes Vazquez

(L-19) Question 17. Given the matrix $\mathbf{A} = \begin{pmatrix} a & 3/5 \\ b & 4/5 \end{pmatrix}$, compute the values (if they exist) of a and b such that

(a) (0.5^{pts}) The matrix $\mathbf{A}$ is orthonormal.
(b) (0.5^{pts}) The columns of $\mathbf{A}$ are linearly independent.
(c) (0.5^{pts}) $\lambda = 0$ is an eigenvalue of $\mathbf{A}$.
(d) (0.5^{pts}) $\mathbf{A}$ is symmetric and negative definite.

(L-19) Question 18.

(a) Obtain the matrix **Q** associated with the quadratic form $q(x, y, z) = x^2 + 2xy + ay^2 + 8z^2$ and classify the matrix **Q** (that is, indicate in which cases the matrix is indefinite, definite, or semidefinite, either positive or negative) for the case when a is equal to one ($a = 1$).

(b) Classify the matrix **Q** when $a \neq 1$.

End of Questions of the Lecture 19

OPTIONAL LECTURE II: Review execises

Questions of the Optional Lecture 2

(L-Opt-2) Question 1. Consider the following matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & a & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

(a) (0.5^{pts}) Show that $\mathbf{A}$ is invertible if and only if $a \neq 0$.
(b) (0.5^{pts}) Is $\mathbf{A}$ positive definite when $a = 1$? Justify your answer.
(c) (1^{pts}) Compute $\mathbf{A}^{-1}$ when $a = 2$.
(d) (0.5^{pts}) How many variables can be chosen as exogenous (or free) variables in the system $\mathbf{A}\boldsymbol{x} = \mathbf{0}$ when $a = 0$? Which ones?

(L-Opt-2) Question 2. True or false (to receive full credit you must justify your answers clearly and concisely)
(a) If $\mathbf{A}$ is symmetric, then so is $\mathbf{A}^2$.
(b) If $\mathbf{A}^2 = \mathbf{A}$ then $(\mathbf{I} - \mathbf{A})^2 = (\mathbf{I} - \mathbf{A})$ where $\mathbf{I}$ is the identity matrix.
(c) If $\lambda = 0$ is an eigenvalue of the squared matrix $\mathbf{A}$, then the linear system $\mathbf{A}\boldsymbol{x} = \mathbf{0}$ is is always solvable and has only one solution.
(d) If $\lambda = 0$ is an eigenvalue of the squared matrix $\mathbf{A}$, then the linear system $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ may be inconsistent.
(e) If a matrix is orthogonal (columns are orthogonal and of unit length), then its inverse is also orthogonal.
(f) If 1 is the only eigenvalue of a 2×2 matrix $\mathbf{A}$, then $\mathbf{A}$ must be the identity matrix $\mathbf{I}$.

(L-Opt-2) Question 3. fill in the blanks, or respond **T**rue/**F**alse.
(a) Any generating system of a vector space contains a basis of the space (T/F)

_____.
(b) Every vector space is finite dimensional (T/F) _____.
(c) That $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n$ are linearly independent means that

_____.
(d) The set that contains only the vector $\mathbf{0}$ is a linearly independent set. (T/F)

_____.
(e) A square matrix of order n by n is diagonalizable when:

_____.
(f) If $\boldsymbol{u} = (1, 2, -1, 1)$, then $||\boldsymbol{u}|| =$ _____.
(g) If $\boldsymbol{u} = (1, 2, -1, 1)$ and $\boldsymbol{v} = (-2, 1, 0, 0)$, then $\boldsymbol{u} \cdot \boldsymbol{v} =$ _____.

(L-Opt-2) Question 4. In the following questions, $\mathbf{A}$ and $\mathbf{B}$ are $n \times n$ matrices. Indicate whether the following statements are true or false (include a brief explanation or a counterexample that justifies your answer):
(a) If $\mathbf{A}$ is not zero, then $\det(\mathbf{A}) \neq 0$.
(b) If $\det(\mathbf{AB}) \neq 0$, then $\mathbf{A}$ is invertible.
(c) If I swap the first two rows of $\mathbf{A}$, its eigenvalues change.
(d) If $\mathbf{A}$ is real and symmetric, then its eigenvalues are real **(no justification is needed here)**.
(e) If the reduced echelon form of $(\mathbf{A} - 5\mathbf{I})$ is the identity matrix, then 5 is not an eigenvalue of $\mathbf{A}$.
(f) Let $\boldsymbol{b}$ a column vector in $\mathbb{R}^n$. If the system $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ has no solution, then $\det(\mathbf{A}) \neq 0$
(g) Let $\mathbf{C}$ of size 3×5. The rank of $\mathbf{C}$ can be 4.
(h) Let $\mathbf{C}$ of size $n \times m$, and $\boldsymbol{b}$ a column vector in $\mathbb{R}^n$. If $\mathbf{C}\boldsymbol{x} = \boldsymbol{b}$ has no solution, then $\operatorname{rg}(\mathbf{C}) < n$.
(i) Every diagonalizable matrix is invertible.
(j) If $\mathbf{A}$ is invertible, then su forma reducida de echelon is the identity matrix.

(L-Opt-2) Question 5. Sean Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 4 \\ 0 & 0 & 5 \end{bmatrix}; \quad \mathbf{B} = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 1 & 3 \\ 0 & 0 & 2 \end{bmatrix}; \quad \mathbf{C} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 0 & 4 \\ 0 & 0 & 6 \end{bmatrix}; \quad \mathbf{D} = \begin{bmatrix} 2 & 3 & 0 \\ 3 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The eigenvalues of $\mathbf{B}$ are 0 and 2. Use this information to answer the following questions. For each matrix, provide an explanation. There may be more than one matrix that satisfies the condition:

(a) Which matrices are invertible?
(b) Which matrices have a repeated eigenvalue?
(c) Which matrices have a rank less than three?
(d) Which matrices are diagonalizable?
(e) For which diagonalizable matrices can we find three mutually orthogonal eigenvectors?

(L-Opt-2) Question 6. Consider the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

(a) Find the eigenvalues and eigenvectors of $\mathbf{A}$.
(b) Is $\mathbf{A}$ diagonalizable?
(c) Is it possible to find a matrix $\mathbf{P}$ such that $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{\intercal}$, where $\mathbf{D}$ is a diagonal matrix?
(d) Compute $|\mathbf{A}^{-1}|$.

(L-Opt-2) Question 7. Consider a 3×3 matrix $\mathbf{A}$ with eigenvalues $\lambda_1 = 1$, $\lambda_2 = 2$, and $\lambda_3 = -1$; and let $\boldsymbol{v}_1 = (1, 0, 1)^{\intercal}$ and $\boldsymbol{v}_2 = (1, 1, 1)^{\intercal}$ be the corresponding eigenvectors to λ_1 and λ_2.

(a) Is $\mathbf{A}$ diagonalizable?
(b) Could $\boldsymbol{v}_3 = (-1, 0, -1)^{\intercal}$ be an eigenvector associated with the eigenvalue $\lambda_3 = -1$?
(c) Compute $\mathbf{A}(\boldsymbol{v}_1 - \boldsymbol{v}_2)$.

(L-Opt-2) Question 8.

(a) (0.5^{pts}) Find a homogeneous system $\mathbf{A}\boldsymbol{x} = \mathbf{0}$ whose solution set is

$$\left\{ \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} \in \mathbb{R}^4 \;\middle|\; \exists \alpha, \beta, \gamma \in \mathbb{R} \quad \text{such that} \quad \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} + \gamma \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \right\}$$

(b) (0.5^{pts}) If the characteristic polynomial of the matrix $\mathbf{A}$ is $p(\lambda) = \lambda^5 + 3\lambda^4 - 24\lambda^3 + 28\lambda^2 - 3\lambda + 10$, find the rank of $\mathbf{A}$.

(L-Opt-2) Question 9. Suppose a square and *invertible* matrix $\underset{n \times n}{\mathbf{A}}$.

(a) What are its column space $\mathcal{C}(\mathbf{A})$ and null space $\mathcal{N}(\mathbf{A})$? (do not answer with the definition, indicate which set of vectors composes each space).
(b) Suppose that $\mathbf{A}$ can be factored as $\mathbf{A} = \mathbf{L}\mathbf{U}$:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 7 & 3 & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

Describe the first elimination step in reducing $\mathbf{A}$ to $\mathbf{U}$. How do you know that $\mathbf{U}$ is also an invertible matrix? What is the value of the determinant of $\mathbf{A}$?
(c) Find a particular 3×3 invertible matrix $\mathbf{A}$ that cannot be factored in the form $\mathbf{L}\mathbf{U}$ (without first permuting the rows). What factorization is still possible in your example? (it is not necessary to perform the factorization). How do you know that your matrix $\mathbf{A}$ is invertible?

End of Questions of the Optional Lecture 2

References

Strang, G. (2003). *Introduction to Linear Algebra*. Wellesley-Cambridge Press, Wellesley, Massachusetts. USA, third ed. ISBN 0-9614088-9-8.

Strang, G. (2006). *Linear algebra and its applications*. Thomson Learning, Inc., fourth ed. ISBN 0-03-010567-6.

Solutions

(L-16) Question 1(a)

$$\begin{bmatrix} -3 & 4 & -4 \\ -3 & 5 & -3 \\ -1 & 2 & 0 \end{bmatrix} \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \\ 0 \end{pmatrix}$$

Therefore, the corresponding eigenvalue of $\boldsymbol{v}$ is -1.

$$\begin{bmatrix} -3 & 4 & -4 \\ -3 & 5 & -3 \\ -1 & 2 & 0 \end{bmatrix} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

Therefore, the corresponding eigenvalue of $\boldsymbol{w}$ is 1.

□

(L-16) Question 1(b) We need to find a vector in the null space of $(\mathbf{A} - 2\mathbf{I})$, in other words, we need to find a solution to

$$\begin{bmatrix} -5 & 4 & -4 \\ -3 & 3 & -3 \\ -1 & 2 & -2 \end{bmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

We can find a solution by Gaussian elimination, but as the third column equals minus the second one, it is easy to realize that one solution is:

$$\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

□

(L-16) Question 2(a) The eigenvalues of

$$\mathbf{A} = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$$

are the solutions to the characteristic equation

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} 3-\lambda & 4 \\ 4 & -3-\lambda \end{vmatrix} = (3-\lambda)(-3-\lambda) - 16 = 0; \quad \Rightarrow \quad \lambda_1 = 5;\ \lambda_2 = -5.$$

We can also use

$$\begin{cases} \lambda_1 \cdot \lambda_2 = & \det \mathbf{A} = -25 \\ \lambda_1 + \lambda_2 = & \operatorname{tr} \mathbf{A} = 0 \end{cases}$$

with the same result.

For $\lambda_1 = 5$, the null space of

$$(\mathbf{A} - \lambda_1\mathbf{I}) = \begin{bmatrix} 3-5 & 4 \\ 4 & -3-5 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ 4 & -8 \end{bmatrix}$$

consists of all multiples of $\boldsymbol{x}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$. Hence, the eigenvectors are the non-null multiples of $\boldsymbol{x}_1$

For $\lambda_2 = -5$, the null space of

$$(\mathbf{A} - \lambda_2\mathbf{I}) = \begin{bmatrix} 3+5 & 4 \\ 4 & -3+5 \end{bmatrix} = \begin{bmatrix} 8 & 4 \\ 4 & 2 \end{bmatrix}$$

consists of all multiples of $\boldsymbol{x}_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$. Hence, the eigenvectors are the non-null multiples of $\boldsymbol{x}_2$.

□

(L-16) Question 2(b) The eigenvalues of

$$\mathbf{B} = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$$

are the solutions to the characteristic equation

$$\det(\mathbf{B}-\lambda\mathbf{I}) = \begin{vmatrix} a-\lambda & b \\ b & a-\lambda \end{vmatrix} = (a-\lambda)^2 - b^2 = \lambda^2 - 2a\lambda + (a^2-b^2) = 0.$$

Therefore

$$\lambda = \frac{2a \pm \sqrt{4a^2 - 4(a^2-b^2)}}{2} = a \pm b.$$

For $\lambda_1 = a+b$, the null space of

$$(\mathbf{B}-\lambda_1\mathbf{I}) = \begin{bmatrix} -b & b \\ b & -b \end{bmatrix}$$

consists of all multiples of $\boldsymbol{x}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Hence, the eigenvectors are the non-null multiples of $\boldsymbol{x}_1$.

For $\lambda_2 = a-b$, the null space of

$$(\mathbf{B}-\lambda_2\mathbf{I}) = \begin{bmatrix} b & b \\ b & b \end{bmatrix}$$

consists of all multiples of $\boldsymbol{x}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$. Hence, the eigenvectors are the non-null multiples of $\boldsymbol{x}_2$.

□

(L-16) Question 3. The numbers on the diagonal of $\mathbf{A}$: 1, 2, 3, 7, 8, and 9.

□

(L-16) Question 4(a) Since the matrix is triangular, the elements on the diagonal are its eigenvalues. For $\lambda_1 = 3$ we need to compute a basis for the null space of $(\mathbf{A}-3\mathbf{I})$.

$$\left[\begin{array}{c}\mathbf{A}-3\mathbf{I}\\ \hline \mathbf{I}\end{array}\right] = \left[\begin{array}{ccc} 0 & 4 & 2 \\ 0 & -2 & 2 \\ 0 & 0 & -3 \\ \hline 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow{\tau\left[\left(\frac{-1}{2}\right)\mathbf{2}+\mathbf{3}\right]} \left[\begin{array}{ccc} 0 & 4 & 0 \\ 0 & -2 & 3 \\ 0 & 0 & -3 \\ \hline 1 & 0 & 0 \\ 0 & 1 & -1/2 \\ 0 & 0 & 1 \end{array}\right].$$

Eigenvector: the non-zero multiples of $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

For $\lambda_2 = 1$ we need to compute a basis for the null space of $(\mathbf{A}-\mathbf{I})$.

$$\left[\begin{array}{c}\mathbf{A}-\mathbf{I}\\ \hline \mathbf{I}\end{array}\right] = \left[\begin{array}{ccc} 2 & 4 & 2 \\ 0 & 0 & 2 \\ 0 & 0 & -1 \\ \hline 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[\left[(-1)\mathbf{1}+\mathbf{3}\right]]{\tau\left[(-2)\mathbf{1}+\mathbf{2}\right]} \left[\begin{array}{ccc} 2 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & -1 \\ \hline 1 & -2 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right].$$

Eigenvector: the non-zero multiples of $\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$.

For $\lambda_3 = 0$ we need to compute a basis for the null space of $(\mathbf{A}-0\mathbf{I})$.

$$\left[\begin{array}{c}\mathbf{A}-0\mathbf{I}\\ \hline \mathbf{I}\end{array}\right] = \left[\begin{array}{ccc} 3 & 4 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ \hline 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow[\left[(-4)\mathbf{1}+\mathbf{2}\right]]{\tau\left[(3)\mathbf{2}\right]} \left[\begin{array}{ccc} 3 & 0 & 0 \\ 0 & 3 & 6 \\ 0 & 0 & 0 \\ \hline 1 & -4 & -2 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{array}\right] \xrightarrow{\tau\left[(-2)\mathbf{2}+\mathbf{3}\right]} \left[\begin{array}{ccc} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \\ \hline 1 & -4 & 6 \\ 0 & 3 & -6 \\ 0 & 0 & 3 \end{array}\right] = \left[\begin{array}{c}\mathbf{L}\\ \hline \mathbf{E}\end{array}\right]$$

The eigenvectors are non-null multiples of $\begin{pmatrix} 6 \\ -6 \\ 3 \end{pmatrix}$.

In addition $\lambda_1 + \lambda_2 + \lambda_3 = 3 + 1 + 0 = 4 = \operatorname{tr}(\mathbf{A})$ and $\lambda_1 \cdot \lambda_2 \cdot \lambda_3 = 3 \cdot 1 \cdot 0 = 0 = |\mathbf{A}|$.

```
A =  Matrix([[3,4,2],[0,1,2],[0,0,0]])
espectro = {3,1,0}
```

```
for l in espectro:
    display(l)
    L   = Elim(  (A-l*I(3)).apila(I(3),1) ,1)
    cL0 = tuple([c+1 for c,i in enumerate( (1,2,3)|L ) if i.es_nulo()])
    display(((4,5,6)|L|cL0).sis())
```

□

(L-16) Question 4(b) The characteristic equation is

$$\begin{vmatrix} -\lambda & 0 & 2 \\ 0 & 2-\lambda & 0 \\ 2 & 0 & -\lambda \end{vmatrix} = (\lambda^2)(2-\lambda) - 4(2-\lambda) = 0$$

It is clear that $\lambda_1 = 2$ is one eigenvalue.

Dividing the characteristic equation by $(2-\lambda)$ we get $\lambda^2 = 4$; therefore $\lambda_2 = 2$ and $\lambda_3 = -2$

For $\lambda = 2$ we need to compute a basis for the null space of

$$\left(\mathbf{A} - 2\mathbf{I}\right) = \begin{bmatrix} -2 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & -2 \end{bmatrix}$$

We get two independent eigenvectors for $\lambda = 2$ (a basis for the null space of $\left(\mathbf{A} - 2\mathbf{I}\right)$):

$$\text{Eigenvectors: all non-null linear combinations of } \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}; \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

For $\lambda_3 = -2$ we need to compute a basis for the null space of

$$\left(\mathbf{A} + 2\mathbf{I}\right) = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 2 \end{bmatrix}$$

$$\text{The corresponding eigenvectors are all non-null multiples of } \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

In addition $\lambda_1 + \lambda_2 + \lambda_3 = 2 + 2 - 2 = 2 = \operatorname{tr}\left(\mathbf{A}\right)$ and $\lambda_1 \cdot \lambda_2 \cdot \lambda_3 = 2 \cdot 2 \cdot (-2) = -8 = |\mathbf{A}|$.

```
B = Matrix([[0,0,2],[0,2,0],[2,0,0]])
l = sympy.symbols('\lambda')
d = Determinante( (B-l*I(3)) ,1)
p = sympy.poly(d.valor)
e = sympy.real_roots(p)
display(Sistema([d, e]))
for i in set(e):
    caso = "\lambda = %d\n" % i
    display(Math(caso))
    L   = Elim(  (B-i*I(3)).apila(I(3),1),1 )
    cL0 = tuple([c+1 for c,v in enumerate((1,2,3)|L) if v.es_nulo()])
    display( ((4,5,6)|L|cL0).sis() )
```

□

(L-16) Question 5(a) Transposing a matrix does not affect the determinant; therefore

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \det\left((\mathbf{A} - \lambda\mathbf{I})^{\mathsf{T}}\right) = \det(\mathbf{A}^{\mathsf{T}} - \lambda\mathbf{I})$$

□

(L-16) Question 5(b) The eigenvalues of the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$$

are the solutions to the characteristic equation

$$\det(\mathbf{A}-\lambda\mathbf{I}) = \begin{vmatrix} 1-\lambda & 2 \\ 0 & 3-\lambda \end{vmatrix} = (1-\lambda)(3-\lambda) = 0;$$

These are equal to the diagonal elements (note that the eigenvalues of an upper (or lower) triangular matrix are always the elements on the diagonal).

For $\lambda_1 = 1$, the null space of

$$(\mathbf{A}-\lambda_1\mathbf{I}) = \begin{bmatrix} 1-1 & 2 \\ 0 & 3-1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 0 & 2 \end{bmatrix}$$

consists of all multiples of the eigenvector $\boldsymbol{x}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

For $\lambda_2 = 3$, the null space of

$$(\mathbf{A}-\lambda_2\mathbf{I}) = \begin{bmatrix} 1-3 & 2 \\ 0 & 3-3 \end{bmatrix} = \begin{bmatrix} -2 & 2 \\ 0 & 0 \end{bmatrix}$$

consists of all multiples of the eigenvector $\boldsymbol{x}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

However, for the transposed matrix

$$\mathbf{A}^\intercal = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix},$$

with the same eigenvalues (same diagonal), we find that

For $\lambda_1 = 1$, the null space of

$$(\mathbf{A}^\intercal-\lambda\mathbf{I}) = \begin{bmatrix} 0 & 0 \\ 2 & 2 \end{bmatrix}$$

consists of all multiples of the eigenvector $\boldsymbol{x}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$.

For $\lambda_2 = 3$, the null space of

$$(\mathbf{A}^\intercal-\lambda\mathbf{I}) = \begin{bmatrix} -2 & 0 \\ 2 & 0 \end{bmatrix}$$

consists of all multiples of the eigenvector $\boldsymbol{x}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Therefore, the eigenvectors of $\mathbf{A}$ and $\mathbf{A}^\intercal$ are different.

□

(L-16) Question 6.

$$\begin{aligned} \mathbf{A}\boldsymbol{x} =& \big(\mathbf{B}+\alpha\mathbf{I}\big)\boldsymbol{x} \\ =& \mathbf{B}\boldsymbol{x} + \alpha\mathbf{I}\boldsymbol{x} \\ =& \lambda\boldsymbol{x} + \alpha\boldsymbol{x} \\ =& (\lambda+\alpha)\boldsymbol{x}; \end{aligned}$$

therefore, $\boldsymbol{x}$ is also an eigenvector of $\mathbf{A}$ associated with the eigenvalue $(\lambda+\alpha)$.

□

(L-16) Question 7(a) Primero calculemos los autovalores:

$$\begin{vmatrix} 1-\lambda & -1 \\ 2 & 4-\lambda \end{vmatrix} = (1-\lambda)(4-\lambda)+2 = \lambda^2-5\lambda+6 = 0; \quad \Rightarrow \quad \begin{cases} \lambda_1 &= 2 \\ \lambda_2 &= 3 \end{cases}$$

Y ahora los autovectores.

Para $\lambda_1 = 2$ buscamos una base del espacio nulo de la matriz

$$\begin{bmatrix} 1-2 & -1 \\ 2 & 4-2 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 2 & 2 \end{bmatrix}$$

Puesto que las dos columnas son iguales, los autovectores son los múltiplos no nulos de $\boldsymbol{x}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$.

Para $\lambda_2 = 3$ buscamos una base del espacio nulo de la matriz

$$\begin{bmatrix} 1-3 & -1 \\ 2 & 4-3 \end{bmatrix} = \begin{bmatrix} -2 & -1 \\ 2 & 1 \end{bmatrix}$$

Puesto que la primera columna es el doble de la segunda, los correspondientes autovectores son los múltiplos no nulos de $\boldsymbol{x}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$

Por último, el determinante de $\mathbf{A}$ es 6, que es igual a 2×3, y la traza es 5, que es igual a $2 + 3$.

□

(L-16) Question 7(b) Calculemos los autovalores:

$$\begin{vmatrix} -6-\lambda & -1 \\ 2 & -3-\lambda \end{vmatrix} = (-6 - \lambda)(-3 - \lambda) + 2 = \lambda^2 + 9\lambda + 20 = 0; \quad \Rightarrow \quad \begin{cases} \lambda_1 = -5 \\ \lambda_2 = -4 \end{cases}.$$

Y ahora los autovectores.

Para $\lambda_1 = -5$ buscamos una base del espacio nulo de la matriz

$$\begin{bmatrix} -6-(-5) & -1 \\ 2 & -3-(-5) \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 2 & 2 \end{bmatrix}.$$

Al igual que en el apartado anterior, los correspondientes autovectores son los múltiplos no nulos de $\boldsymbol{x}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$.

Para $\lambda_2 = -4$ buscamos una base del espacio nulo de la matriz

$$\begin{bmatrix} -6-(-4) & -1 \\ 2 & -3-(-4) \end{bmatrix} = \begin{bmatrix} -2 & -1 \\ 2 & 1 \end{bmatrix}$$

De manera idéntica al apartado (a), los correspondientes autovectores son los múltiplos no nulos de $\boldsymbol{x}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$.

Así pues, tras restar 7 veces la matriz identidad ($\mathbf{B} = \mathbf{A} - 7\mathbf{I}$), los autovalores son los de la matriz original menos 7; es decir, $\lambda_1 = 2 - 7 = -5$ y $\lambda_2 = 3 - 7 = -4$; y los autovectores son idénticos a los de la matriz original $\mathbf{A}$.

□

(L-16) Question 8(a) Let $\boldsymbol{x}$ be a vector such that $\mathbf{A}\boldsymbol{x} = \lambda\boldsymbol{x}$; then

$$\mathbf{B}\boldsymbol{x} = (\mathbf{A} - 7\mathbf{I})\boldsymbol{x} = \mathbf{A}\boldsymbol{x} - 7\boldsymbol{x} = (\lambda - 7)\boldsymbol{x}$$

Therefore, $\boldsymbol{x}$ is also an eigenvector of $\mathbf{B}$ with an associated eigenvalue of $(\lambda - 7)$.

□

(L-16) Question 8(b)

$$\begin{aligned} \mathbf{A}\boldsymbol{x} &= \lambda\boldsymbol{x} \\ \boldsymbol{x} &= \lambda(\mathbf{A}^{-1})\boldsymbol{x} \\ \frac{1}{\lambda}\boldsymbol{x} &= (\mathbf{A}^{-1})\boldsymbol{x} \end{aligned}$$

The last equality $(\mathbf{A}^{-1})\boldsymbol{x} = (1/\lambda)\boldsymbol{x}$ implies that $\boldsymbol{x}$ is also an eigenvector of $\mathbf{A}^{-1}$ with an associated eigenvalue equal to $1/\lambda$.

□

(L-16) Question 9. If λ is an eigenvalue of $\mathbf{A}$, then $\mathbf{A}\boldsymbol{x} = \lambda\boldsymbol{x}$. Then

$$\mathbf{A}^2\boldsymbol{x} = \mathbf{A}\mathbf{A}\boldsymbol{x} = \mathbf{A}(\lambda\boldsymbol{x}) = \lambda\mathbf{A}\boldsymbol{x} = \lambda^2\boldsymbol{x}.$$

However, since $\mathbf{A}^2 = \mathbf{A}$ it follows that

$$\mathbf{A}^2\boldsymbol{x} = \mathbf{A}\boldsymbol{x} = \lambda^2\boldsymbol{x} = \lambda\boldsymbol{x},$$

therefore

$$\lambda^2 = \lambda.$$

The only two possible values are either 0 or 1.

□

(L-16) Question 10.

$$\mathbf{A}(\boldsymbol{v}_1 + \boldsymbol{v}_2 - \boldsymbol{v}_3) = \mathbf{A}\boldsymbol{v}_1 + \mathbf{A}\boldsymbol{v}_2 - \mathbf{A}\boldsymbol{v}_3 = \boldsymbol{v}_1 + 2\boldsymbol{v}_2 - 3\boldsymbol{v}_3.$$

□

(L-16) Question 11. The characteristic equation of the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

is $(1-\lambda)^2 - 1 = 0$. Thus, the matrix has eigenvalues $\lambda = 0$ and $\lambda = 2$.

However, the characteristic equation of its row echelon form is

$$\mathbf{L} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

is $(1-\lambda)(-\lambda) = 0$. Thus, the new eigenvalues are $\lambda = 0$ and $\lambda = 1$.

The zero eigenvalues do not change. There are as many as the number of columns minus the rank of the matrix (that is, as many as the number of free columns); and neither the number of columns nor the rank of the matrix changes when applying elementary transformations. Therefore, the number of zero eigenvalues remains the same after applying the elimination method.

□

(L-16) Question 12. It suffices to set λ to zero.

□

(L-16) Question 13. For $\mathbf{A}$, the sum of λ_1 and λ_2 is -1 (the trace) and the product is -6 (the determinant), so $\lambda_1 = -3$ and $\lambda_2 = 2$

For $\lambda_1 = -3$; a basis for the null space of $(\mathbf{A}+3\mathbf{I}) = \begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix}$ is $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$. Its non-zero multiples are the corresponding eigenvectors.

And for $\lambda_2 = 2$; a basis for the null space of $(\mathbf{A} - 2\mathbf{I}) = \begin{bmatrix} -3 & 3 \\ 2 & -2 \end{bmatrix}$ is $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Its non-zero multiples are the corresponding eigenvectors.

On the other hand, for $\mathbf{A}^2$, the sum of λ_1 and λ_2 is 13 (the trace) and the product is 36 (the determinant), so $\lambda_1 = 9$ and $\lambda_2 = 4$

For $\lambda_1 = 9$; a basis for the null space of $(\mathbf{A} - 9\mathbf{I}) = \begin{bmatrix} -2 & -3 \\ -2 & -3 \end{bmatrix}$ is $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$. Its non-zero multiples are the corresponding eigenvectors.

And for $\lambda_2 = 4$; a basis for the null space of $(\mathbf{A} - 4\mathbf{I}) = \begin{bmatrix} 3 & -3 \\ -2 & 2 \end{bmatrix}$ is $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Its non-zero multiples are the corresponding eigenvectors.

So $\mathbf{A}^2$ has the same **eigenvectors** as $\mathbf{A}$. If the eigenvalues of $\mathbf{A}$ are λ_1 and λ_2, the eigenvalues of $\mathbf{A}^2$ are **The square of the above**: (λ_1^2 y λ_2^2).

□

(L-16) Question 14. The eigenvalues of $\mathbf{A}^2$ are the square of the eigenvalues of $\mathbf{A}$, so

$$\operatorname{tr}\left(\mathbf{A}^2\right) = 1^2 + 2^2 + 4^2 = 21$$

Reasoning in the same way

$$\left|(\mathbf{A}^{-1})^{\intercal}\right| = \det \mathbf{A}^{-1} = (1^{-1})(2^{-1})(4^{-1}) = 1 \cdot \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}.$$

□

(L-16) Question 15. The condition states that $(\mathbf{A}^2 - 4\mathbf{I})$ is singular. However, we know that if $\lambda_1, \dots \lambda_n$ are eigenvalues of $\mathbf{A}$, then the eigenvalues of $(\mathbf{A}^2 - 4\mathbf{I})$ are $\lambda_1^2 - 4, \lambda_1^2 - 4, \dots, \lambda_n^2 - 4$. The condition that $(\mathbf{A}^2 - 4\mathbf{I})$ is singular states that one of $\lambda_i^2 - 4$ is zero, and therefore $\lambda_i = 2$ o $\lambda_i = -2$. That is, $\mathbf{A}$ has an eigenvalue of 2 or -2.

□

(L-16) Question 16. First, the last two columns of $\mathbf{A}$ are identical. Thus, $\mathbf{A}$ is singular and must have an eigenvalue $\lambda_1 = 0$. Additionally, we observe that $\mathbf{A}$ is a Markov matrix. This means that $\lambda_2 = 1$ is an eigenvalue of $\mathbf{A}$. Finally, we know that the trace of $\mathbf{A}$ is the sum of its three eigenvalues. Therefore, $\operatorname{tr}(\mathbf{A}) = 0.5 + 0.5 + 0.3 = 1.3$ and the last eigenvalue is $\lambda_3 = 1.3 - 1 - 0 = 0.3$.

□

(L-17) Question 1(a) From QUESTION 5 on page 7 (L-19) we have already calculated the eigenvalues and eigenvectors of $\mathbf{A}$, therefore we know

$$\mathbf{S} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}; \qquad \mathbf{D} = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}.$$

We need to compute $\mathbf{S}^{-1}$:

$$\left[\begin{array}{c} \mathbf{S} \\ \hline \mathbf{I} \end{array}\right] = \left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \\ \hline 1 & 0 \\ 0 & 1 \end{array}\right] \xrightarrow[[(-1)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \\ \hline 1 & -1 \\ 0 & 1 \end{array}\right] = \left[\begin{array}{c} \mathbf{I} \\ \hline \mathbf{S}^{-1} \end{array}\right]$$

Therefore

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}.$$

O aplicando el algoritmo de diagonalización por bloques (que en este caso sabemos que arrojará una matriz diagonal, puesto que los autovalores no se repiten)

$$\left[\begin{array}{c} \mathbf{A} \\ \hline \mathbf{I} \end{array}\right] \xrightarrow[3\mathbf{I}]{(-)} \left[\begin{array}{cc} -2 & 2 \\ 0 & 0 \\ \hline 1 & 0 \\ 0 & 1 \end{array}\right] \xrightarrow[[(1)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \left[\begin{array}{cc} -2 & 0 \\ 0 & 0 \\ \hline 1 & 1 \\ 0 & 1 \end{array}\right] \xrightarrow[[(-1)\mathbf{2}+\mathbf{1}]]{\boldsymbol{\tau}} \left[\begin{array}{cc} -2 & 0 \\ 0 & 0 \\ \hline 1 & 1 \\ 0 & 1 \end{array}\right] \xrightarrow[3\mathbf{I}]{(+)} \left[\begin{array}{cc} 1 & 0 \\ 0 & 3 \\ \hline 1 & 1 \\ 0 & 1 \end{array}\right] \xrightarrow[1\mathbf{I}]{(-)} \left[\begin{array}{cc} 0 & 0 \\ 0 & 2 \\ \hline 1 & 1 \\ 0 & 1 \end{array}\right] \xrightarrow[1\mathbf{I}]{(+)} \left[\begin{array}{cc} 1 & 0 \\ 0 & 3 \\ \hline 1 & 1 \\ 0 & 1 \end{array}\right]$$

```
A = Matrix([[1,2],[0,3]])
lambdas = [1,3]
D = DiagonalizaS(A,lambdas,1)
display(D)
display(D.S)
```

□

(L-17) Question 1(b) First we will solve the characteristic equation $\det(\mathbf{B} - \lambda\mathbf{I}) = 0$ to find the eigenvalues

$$\begin{vmatrix} 1-\lambda & 1 \\ 2 & 2-\lambda \end{vmatrix} = (1-\lambda)(2-\lambda) - 2 = \lambda^2 - 3\lambda = 0; \quad \Rightarrow \quad \begin{cases} \lambda_1 & = 0 \\ \lambda_2 & = 3 \end{cases}$$

(Why could we know that one eigenvalue is zero before solving the characteristic equation?)

- for $\lambda_1 = 0$, the null space of $(\mathbf{B} - 0\mathbf{I}) = \mathbf{B} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$ consists of all multiples of the eigenvector

$$\boldsymbol{x}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

- for $\lambda_2 = 3$, the null space of $(\mathbf{B} - \lambda\mathbf{I}) = \begin{bmatrix} -2 & 1 \\ 2 & -1 \end{bmatrix}$ consists of all the multiples of

$$\boldsymbol{x}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Therefore:

$$\mathbf{S} = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}; \qquad \mathbf{D} = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}.$$

O aplicando el algoritmo de diagonalización por bloques (que en este caso sabemos que arrojará una matriz diagonal, puesto que los autovalores no se repiten)

$$\left[\begin{array}{c}\mathbf{A}\\\hline \mathbf{I}\end{array}\right] \xrightarrow[0\mathbf{I}]{(-)} \left[\begin{array}{cc} 1 & 1 \\ 2 & 2 \\ \hline 1 & 0 \\ 0 & 1 \end{array}\right] \xrightarrow[[(-1)\mathbf{1}+\mathbf{2}]]{\tau} \left[\begin{array}{cc} 1 & 0 \\ 2 & 0 \\ \hline 1 & -1 \\ 0 & 1 \end{array}\right] \xrightarrow[[(1)\mathbf{2}+\mathbf{1}]]{\tau} \left[\begin{array}{cc} 3 & 0 \\ 2 & 0 \\ \hline 1 & -1 \\ 0 & 1 \end{array}\right] \xrightarrow[0\mathbf{I}]{(+)} \left[\begin{array}{cc} 3 & 0 \\ 2 & 0 \\ \hline 1 & -1 \\ 0 & 1 \end{array}\right]$$

$$\xrightarrow[3\mathbf{I}]{(-)} \left[\begin{array}{cc} 0 & 0 \\ 2 & -3 \\ \hline 1 & -1 \\ 0 & 1 \end{array}\right] \xrightarrow[[(2)\mathbf{2}+\mathbf{1}]]{\tau} \left[\begin{array}{cc} 0 & 0 \\ 0 & -3 \\ \hline 1 & -1 \\ 2 & 1 \end{array}\right] \xrightarrow[[(-2)\mathbf{1}+\mathbf{2}]]{\tau} \left[\begin{array}{cc} 0 & 0 \\ 0 & -3 \\ \hline 1 & -1 \\ 2 & 1 \end{array}\right] \xrightarrow[3\mathbf{I}]{(+)} \left[\begin{array}{cc} 3 & 0 \\ 0 & 0 \\ \hline 1 & -1 \\ 2 & 1 \end{array}\right] = \left[\begin{array}{c}\mathbf{D}\\\hline \mathbf{S}\end{array}\right]$$

Finally we compute $\mathbf{S}^{-1}$:

$$\left[\begin{array}{c}\mathbf{S}\\\hline \mathbf{I}\end{array}\right] = \left[\begin{array}{cc} 1 & -1 \\ 2 & 1 \\ \hline 1 & 0 \\ 0 & 1 \end{array}\right] \xrightarrow[[(1)\mathbf{1}+\mathbf{2}]]{\tau} \left[\begin{array}{cc} 1 & 0 \\ 2 & 3 \\ \hline 1 & 1 \\ 0 & 1 \end{array}\right] \xrightarrow[[(-2)\mathbf{2}+\mathbf{1}]]{\tau} \left[\begin{array}{cc} 3 & 0 \\ 0 & 3 \\ \hline 1 & 1 \\ -2 & 1 \end{array}\right] \xrightarrow[[(\frac{1}{3})\mathbf{2}]]{\tau} \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \\ \hline \frac{1}{3} & \frac{1}{3} \\ \frac{-2}{3} & \frac{1}{3} \end{array}\right] = \left[\begin{array}{c}\mathbf{I}\\\hline \mathbf{S}^{-1}\end{array}\right]$$

The factorization is

$$\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1/3 & 1/3 \\ -2/3 & 1/3 \end{bmatrix} = \mathbf{SDS}^{-1}.$$

```
B = Matrix([[1,1],[2,2]])
l = sympy.symbols('\lambda')
d = Determinante( (B-l*I(B.n)) ,1)
p = sympy.poly(d.valor)
e = sympy.real_roots(p)
display(Sistema([d, e]))
S = Sistema([])
D = tuple()
for i in set(e):
    caso = "\lambda = %d\n" % i
    display(Math(caso))
    L   = Elim( (B-i*I(B.n)).apila(I(B.n),1), 1)
    cL0 = tuple([c+1 for c,v in enumerate(slice(1,B.n)|L) if v.es_nulo()])
    S = S.concatena(slice(B.n+1,None)|L|cL0).sis()
    D = D+(i,)*e.count(i)

S = Matrix(S)
D = Vector(D).diag()
display(Sistema([D,S,S**-1]))
display(Sistema([(S**-1)*B*S, S*D*(S**-1)]))
```

□

(L-17) Question 2(a)

$$\begin{cases} \lambda_1 + \lambda_2 = 0 \\ \lambda_1 \cdot \lambda_2 = 0 \end{cases} \quad \Rightarrow \quad \lambda_1 = \lambda_2 = 0.$$

This matrix can only be diagonalized if the null space of $(\mathbf{A}_1 - 0\mathbf{I}) = \mathbf{A}_1$ has dimension 2. Since $\mathbf{A}_1$ will only have one free column $\dim \mathcal{N}\left(\mathbf{A}\right) = 1$; therefore this matrix cannot be diagonalized.

□

(L-17) Question 2(b) This matrix has no repeated eigenvalues; therefore it can be diagonalized.

□

(L-17) Question 2(c) The eigenvalue $\lambda = 2$ is repeated; therefore, for the matrix to be diagonalizable it is necessary that the null space of $(\mathbf{A}_3 - 2\mathbf{I})$ has dimension 2 (two linearly independent eigenvectors), but it only has dimension 1. This matrix cannot be diagonalized.

□

(L-17) Question 3. Characteristic equation: $(4-\lambda)(2-\lambda) - 3 = \lambda^2 - 6\lambda + 5 = 0; \quad \Rightarrow \quad \lambda_1 = 5; \quad \lambda_2 = 1.$

$$\lambda_1 = 5 \quad \Longrightarrow \quad (\mathbf{A} - 5\mathbf{I}) = \begin{bmatrix} -1 & 3 \\ 1 & -3 \end{bmatrix} \quad \Rightarrow \quad \boldsymbol{x} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$\lambda_1 = 1 \quad \Longrightarrow \quad (\mathbf{A} - \mathbf{I}) = \begin{bmatrix} 3 & 3 \\ 1 & 1 \end{bmatrix} \quad \Rightarrow \quad \boldsymbol{x} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Alternatively:

$$\begin{bmatrix} \mathbf{A} \\ \hline \mathbf{I} \end{bmatrix} \xrightarrow[1\mathbf{I}]{(-)} \left[\begin{array}{cc} 3 & 3 \\ 1 & 1 \\ \hline 1 & 0 \\ 0 & 1 \end{array}\right] \xrightarrow[[(-1)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \left[\begin{array}{cc} 3 & 0 \\ 1 & 0 \\ \hline 1 & -1 \\ 0 & 1 \end{array}\right] \xrightarrow[[(1)\mathbf{2}+\mathbf{1}]]{\boldsymbol{\tau}} \left[\begin{array}{cc} 4 & 0 \\ 1 & 0 \\ \hline 1 & -1 \\ 0 & 1 \end{array}\right] \xrightarrow[1\mathbf{I}]{(+)} \left[\begin{array}{cc} 5 & 0 \\ 1 & 1 \\ \hline 1 & -1 \\ 0 & 1 \end{array}\right]$$

$$\xrightarrow[5\mathbf{I}]{(-)} \left[\begin{array}{cc} 0 & 0 \\ 1 & -4 \\ \hline 1 & -1 \\ 0 & 1 \end{array}\right] \xrightarrow[[(1)\mathbf{2}+\mathbf{1}]]{\substack{\boldsymbol{\tau} \\ [(4)\mathbf{1}]}} \left[\begin{array}{cc} 0 & 0 \\ 0 & -4 \\ \hline 3 & -1 \\ 1 & 1 \end{array}\right] \xrightarrow[[(-1)\mathbf{1}+\mathbf{2}]]{\substack{\boldsymbol{\tau} \\ [(1/4)\mathbf{1}]}} \left[\begin{array}{cc} 0 & 0 \\ 0 & -4 \\ \hline 3 & -1 \\ 1 & 1 \end{array}\right] \xrightarrow[5\mathbf{I}]{(+)} \left[\begin{array}{cc} 5 & 0 \\ 0 & 1 \\ \hline 3 & -1 \\ 1 & 1 \end{array}\right]$$

Hence,

$$\mathbf{A}^{100} = \mathbf{S}\mathbf{D}^{100}\mathbf{S}^{-1} = \begin{bmatrix} 3 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 5^{100} & \\ & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & -1 \end{bmatrix}^{-1} = \begin{bmatrix} 5 \cdot 5^{100} & 1 \\ 5^{100} & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 3 \end{bmatrix} \frac{1}{4} = \frac{1}{4} \begin{bmatrix} 3 \cdot 5^{100} - 1 & 3 \cdot 5^{100} + 3 \\ 5^{100} + 1 & 5^{100} - 3 \end{bmatrix}$$

□

(L-17) Question 4(a) Yes, since it has no eigenvalue equal to zero; in fact, its determinant is $|\mathbf{A}| = 1 \times 1 \times 2 = 2 \neq 0$.

□

(L-17) Question 4(b) We cannot know. It depends on how many linearly independent eigenvectors we can find.

□

(L-17) Question 4(c) We cannot know. It depends on how many linearly independent eigenvectors we can find.

□

(L-17) Question 5(a) Since the matrix is triangular, the eigenvalues are the elements on the main diagonal ($\lambda = 4$ and $\lambda = 2$, both with algebraic multiplicity equal to 2). Thus, for the matrix to be diagonalizable, the rank of the matrix $(\mathbf{A} - \lambda\mathbf{I})$ must be two in both cases. Let's check if this is true.

$$\text{rg}\left(\mathbf{A} - 4\mathbf{I}\right) = \text{rg}\left(\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 1 & 0 & 0 & -2 \end{bmatrix}\right) = 2; \quad \text{rg}\left(\mathbf{A} - 2\mathbf{I}\right) = \text{rg}\left(\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}\right) = 2.$$

Thus, we can conclude that $\mathbf{A}$ is diagonalizable.

Observing the matrix $(\mathbf{A}-4\mathbf{I})$, it is easy to see that two eigenvectors associated with $\lambda = 4$ are $\begin{pmatrix} 2, & 0, & 0, & 1, \end{pmatrix}$ and $\begin{pmatrix} 0, & 1, & 0, & 0, \end{pmatrix}$; And observing the matrix $(\mathbf{A}-2\mathbf{I})$, that two eigenvectors associated with $\lambda = 2$ are $\begin{pmatrix} 0, & 0, & 1, & 0, \end{pmatrix}$ and $\begin{pmatrix} 0, & 0, & 0, & 1, \end{pmatrix}$. Thus, $\mathbf{D} = \begin{bmatrix} 4 & & & \\ & 4 & & \\ & & 2 & \\ & & & 2 \end{bmatrix}$; and $\mathbf{S} = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$.

□

(L-17) Question 5(b) Since we have seen that $\boldsymbol{v}$ is an eigenvector of $\mathbf{A}$ associated with the eigenvalue $\lambda = 2$, we know that $\mathbf{A}\boldsymbol{v} = 2\boldsymbol{v}$, and therefore:

$$\begin{aligned}\left(\mathbf{A}^6\right)\boldsymbol{v} =& \mathbf{A}\cdot\mathbf{A}\cdot\mathbf{A}\cdot\mathbf{A}\cdot\mathbf{A}\cdot\mathbf{A}\cdot\boldsymbol{v}\\ =& \mathbf{A}\cdot\mathbf{A}\cdot\mathbf{A}\cdot\mathbf{A}\cdot\mathbf{A}\cdot\lambda\boldsymbol{v}\\ =& \mathbf{A}\cdot\mathbf{A}\cdot\mathbf{A}\cdot\mathbf{A}\cdot\lambda^2\boldsymbol{v}\\ =& \mathbf{A}\cdot\mathbf{A}\cdot\mathbf{A}\cdot\lambda^3\boldsymbol{v}\\ =& \mathbf{A}\cdot\mathbf{A}\cdot\lambda^4\boldsymbol{v}\\ =& \mathbf{A}\cdot\lambda^5\boldsymbol{v}\\ =& \lambda^6\boldsymbol{v} = 2^6\boldsymbol{v} = 64\boldsymbol{v} = \begin{pmatrix}0, & 0, & 0, & 64,\end{pmatrix}.\end{aligned}$$

□

(L-17) Question 5(c) Since none of the eigenvalues are zero, the matrix is of full rank, i.e., it is invertible.

□

(L-17) Question 5(d) Since $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$, then

$$\mathbf{A}^{-1} = \left(\mathbf{S}\mathbf{D}\mathbf{S}^{-1}\right)^{-1} = \left(\mathbf{D}\mathbf{S}^{-1}\right)^{-1}\mathbf{S}^{-1} = \left(\mathbf{S}^{-1}\right)^{-1}\mathbf{D}^{-1}\mathbf{S}^{-1} = \mathbf{S}\mathbf{D}^{-1}\mathbf{S}^{-1};$$

That is, the eigenvectors $\mathbf{S}$ are the same, but the eigenvalues $\mathbf{D}^{-1}$ are the inverses of the eigenvalues of the matrix $\mathbf{A}$.

□

(L-17) Question 6.

$$\mathbf{A}^3 = (\mathbf{S})(\mathbf{D}^3)(\mathbf{S}^{-1}); \quad \text{y} \quad \mathbf{A}^{-1} = (\mathbf{S})(\mathbf{D}^{-1})(\mathbf{S}^{-1}).$$

□

(L-17) Question 7(a) We need to solve the characteristic equation

$$\begin{bmatrix}1-\lambda & 0\\ -1 & 2-\lambda\end{bmatrix} = (1-\lambda)(2-\lambda) = 0.$$

Thus, $\lambda = 1,\ 2$ (since it is a triangular matrix).

□

(L-17) Question 7(b) To find the eigenvectors corresponding to λ, we need to find the null space of $(\mathbf{A} - \lambda\mathbf{I})$. There are two cases:

- Case $\lambda = 1$. Here we need to solve $\begin{bmatrix}0 & 0\\ -1 & 1\end{bmatrix}\begin{pmatrix}a\\ b\end{pmatrix} = \begin{pmatrix}0\\ 0\end{pmatrix}$. Thus, one eigenvector is $\begin{pmatrix}1\\ 1\end{pmatrix}$.
- Case $\lambda = 2$. Here we need to solve $\begin{bmatrix}-1 & 0\\ -1 & 0\end{bmatrix}\begin{pmatrix}a\\ b\end{pmatrix} = \begin{pmatrix}0\\ 0\end{pmatrix}$. In this case, an eigenvector is $\begin{pmatrix}0\\ 1\end{pmatrix}$.

□

(L-17) Question 7(c) From the previous sections, we conclude that:

$$\mathbf{S} = \begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}; \qquad \mathbf{D} = \begin{bmatrix}1 & 0\\ 0 & 2\end{bmatrix}.$$

Since $\mathbf{S}$ is an elementary matrix, we know that its inverse is:

$$\mathbf{S}^{-1} = \begin{bmatrix}1 & 0\\ -1 & 1\end{bmatrix}$$

Thus

$$\mathbf{A} = \begin{bmatrix}1 & 0\\ -1 & 2\end{bmatrix} = \begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 2\end{bmatrix}\begin{bmatrix}1 & 0\\ -1 & 1\end{bmatrix} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}.$$

However, we could have done it this way

$$\left[\begin{array}{c}\mathbf{A}\\\hline \mathbf{I}\end{array}\right] \xrightarrow[2\mathbf{I}]{(-)} \left[\begin{array}{cc}-1&0\\-1&0\\\hline 1&0\\0&1\end{array}\right] \xrightarrow[2\mathbf{I}]{(+)} \left[\begin{array}{cc}1&0\\-1&2\\\hline 1&0\\0&1\end{array}\right] \xrightarrow[1\mathbf{I}]{(-)} \left[\begin{array}{cc}0&0\\\hline -1&1\\\hline 1&0\\0&1\end{array}\right] \xrightarrow[[(1)\mathbf{2}+\mathbf{1}]]{\boldsymbol{\tau}} \left[\begin{array}{cc}0&0\\\hline 0&1\\\hline 1&0\\1&1\end{array}\right] \xrightarrow[[(-1)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \left[\begin{array}{cc}0&0\\\hline 0&1\\\hline 1&0\\1&1\end{array}\right] \xrightarrow[1\mathbf{I}]{(+)} \left[\begin{array}{cc}1&0\\0&2\\\hline 1&0\\1&1\end{array}\right]$$

□

(L-17) Question 8(a) True. Since none of the eigenvalues is zero, the determinant (which is equal to the product of the eigenvalues) is necessarily non-zero, and therefore the matrix is invertible.

□

(L-17) Question 8(b) False. If there are no repeated eigenvalues, we know that the matrix must be diagonalizable, as there are sufficient linearly independent eigenvectors to generate an invertible matrix $\mathbf{S}$. Since the eigenvalue 2 is repeated, we cannot know if there are enough linearly independent eigenvectors.

□

(L-17) Question 8(c) False. We cannot know. We need to determine if there are three linearly independent eigenvectors (we would know this if there were no repeated eigenvalues).

□

(L-17) Question 9(a)

$$\mathbf{A}_1 = \begin{bmatrix}8 & a\\ b & 2\end{bmatrix}, \quad a = \frac{-9}{b}; \qquad \mathbf{A}_2 = \begin{bmatrix}9 & 4\\ -4 & 1\end{bmatrix}; \qquad \mathbf{A}_3 = \begin{bmatrix}10 & 5\\ -5 & 0\end{bmatrix}$$

□

(L-17) Question 9(b) The null space of the matrix $(\mathbf{A}_1 - 5\mathbf{I})$, that is, of $\begin{bmatrix}3 & 3\\ -3 & -3\end{bmatrix}$; is the set of vectors that are multiples of $\boldsymbol{x} = \begin{pmatrix}-1\\1\end{pmatrix}$, and it is the same as the null space of the matrices $(\mathbf{A}_2 - 5\mathbf{I}) = \begin{bmatrix}4 & 4\\ -4 & -4\end{bmatrix}$ y $(\mathbf{A}_2 - 5\mathbf{I}) = \begin{bmatrix}5 & 5\\ -5 & -5\end{bmatrix}$. Therefore, in all three cases, we cannot find two linearly independent eigenvectors, and as a result, these matrices are not diagonalizable.

□

(L-17) Question 10(a) $\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix} = \begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}2 & 0\\ 0 & 0\end{bmatrix}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}^{-1}$.

□

(L-17) Question 10(b) $\begin{bmatrix}2 & 1\\ 0 & 0\end{bmatrix} = \begin{bmatrix}1 & 1\\ 0 & -2\end{bmatrix}\begin{bmatrix}2 & 0\\ 0 & 0\end{bmatrix}\begin{bmatrix}1 & 1\\ 0 & -2\end{bmatrix}^{-1}$.

□

(L-17) Question 11.

$$\begin{bmatrix}3 & 2\\ 1 & 1\end{bmatrix}^{-1} = \begin{bmatrix}1 & -2\\ -1 & 3\end{bmatrix}; \qquad \begin{bmatrix}3 & 2\\ 1 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 4\end{bmatrix}\begin{bmatrix}1 & -2\\ -1 & 3\end{bmatrix} = \begin{bmatrix}-5 & 18\\ -3 & 10\end{bmatrix}$$

Verification: Trace = 5; determinant = 4.

□

(L-17) Question 12. We know it is diagonalizable since it has no repeated eigenvalues. The eigenvalue matrix $\mathbf{D}$ is a diagonal matrix with the values 1, 2, and 7 on its main diagonal (in any order).

□

(L-17) Question 13(a) The three eigenvalues are equal to 1 (they are the elements on the diagonal since $\mathbf{A}$ is triangular).

The eigenvectors for the only (triple) eigenvalue $\lambda = 1$ are calculated starting from the matrix $[\mathbf{A} - 1\mathbf{I}]$:

$$\mathbf{A} - 1\mathbf{I} = \begin{bmatrix}0 & 0 & 0\\ -2 & 0 & 0\\ 1 & 0 & 0\end{bmatrix}$$

which has only two free columns; therefore, we need to find two special solutions (whose linear combinations form the null space of the previous matrix and which constitute the eigenvectors associated with the eigenvalue $\lambda = 1$):

$$\boldsymbol{x}_a = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}; \qquad \boldsymbol{x}_b = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

□

(L-17) Question 13(b) The matrix is not diagonalizable, as we can find at most 2 linearly independent eigenvectors (to be diagonalizable, we would need to find 3).

□

(L-17) Question 14(a) It is diagonalizable, as it has three linearly independent eigenvectors. Therefore:

$$\mathbf{S} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix}; \qquad \mathbf{D} = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 0 \end{bmatrix}.$$

□

(L-17) Question 14(b)

$$\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 1 & \\ & & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

□

(L-17) Question 15.

$$\left(\mathbf{A}^3\right)\boldsymbol{v} = \begin{bmatrix} 2 & 2 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 2 & \\ & -2 \end{bmatrix}^3 \begin{bmatrix} 2 & 2 \\ 0 & -1 \end{bmatrix}^{-1} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$
$$= \begin{bmatrix} 2 & 2 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 2^3 & \\ & -2^3 \end{bmatrix} \begin{bmatrix} 1/2 & 1 \\ 0 & -1 \end{bmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{bmatrix} 8 & 32 \\ 0 & -8 \end{bmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -24 \\ 8 \end{pmatrix}.$$

□

(L-18) Question 1(a) Characteristic equation:

$$0 = |\mathbf{A} - \lambda \mathbf{I}| = \lambda^2 - \lambda,$$

therefore $\lambda_1 = 0$ and $\lambda_2 = 1$

- $\lambda_1 = 0$

$$(\mathbf{A} - \mathbf{0}) = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

  The corresponding unit eigenvector is $\boldsymbol{x}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

- $\lambda_2 = 1$

$$(\mathbf{A} - \mathbf{I}) = \begin{bmatrix} -1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}$$

  The corresponding unit eigenvector is $\boldsymbol{x}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Therefore

$$\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\mathsf{T}} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & \\ & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \frac{1}{\sqrt{2}}$$

□

(L-18) Question 1(b) Characteristic equation:

$$0 = |\mathbf{B} - \lambda\mathbf{I}| = \lambda^2 - 1,$$

therefore $\lambda = \pm 1$

- $\lambda_1 = 1$

$$(\mathbf{B} - \mathbf{I}) = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

  The corresponding unit eigenvector is $\boldsymbol{x}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

- $\lambda_2 = -1$

$$(\mathbf{B} + \mathbf{I}) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

  The corresponding unit eigenvector is $\boldsymbol{x}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

Therefore

$$\mathbf{B} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\mathsf{T} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & \\ & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \frac{1}{\sqrt{2}}$$

□

(L-18) Question 1(c) Characteristic equation:

$$0 = |\mathbf{C} - \lambda\mathbf{I}| = (3 - \lambda)(-3 - \lambda) - 16 = \lambda^2 - 25,$$

therefore $\lambda = \pm 5$

- $\lambda_1 = 5$

$$(\mathbf{C} - 5\mathbf{I}) = \begin{bmatrix} -2 & 4 \\ 4 & -8 \end{bmatrix}$$

  The corresponding unit eigenvector is $\boldsymbol{x}_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

- $\lambda_2 = -5$

$$(\mathbf{C} + 5\mathbf{I}) = \begin{bmatrix} 8 & 4 \\ 4 & -2 \end{bmatrix}$$

  The corresponding unit eigenvector is $\boldsymbol{x}_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} -1 \\ 2 \end{pmatrix}$

Therefore

$$\mathbf{C} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\mathsf{T} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 5 & \\ & -5 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \frac{1}{\sqrt{5}}$$

□

(L-18) Question 2. Characteristic equation:

$$|\mathbf{A} - \lambda\mathbf{I}| = \begin{vmatrix} 1-\lambda & 1 & 1 \\ 1 & -\lambda & 0 \\ 1 & 0 & -\lambda \end{vmatrix} = -\lambda^3 + \lambda^2 + 2\lambda = 0 \quad \Rightarrow \quad \begin{cases} \lambda = 0 \\ -\lambda^2 + \lambda + 2 \Rightarrow \begin{cases} \lambda = 2 \\ \lambda = -1 \end{cases} \end{cases}$$

$\lambda = 0$, 2, -1; with unit eigenvectors

$$\boldsymbol{x}_1 = \pm\frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}; \qquad \boldsymbol{x}_2 = \pm\frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}; \qquad \boldsymbol{x}_3 = \pm\frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}.$$

since

$$\left[\begin{array}{c}\mathbf{A}-0\mathbf{I}\\ \hline \mathbf{I}\end{array}\right]=\left[\begin{array}{ccc}1&1&1\\1&0&0\\1&0&0\\ \hline 1&0&0\\0&1&0\\0&0&1\end{array}\right]\xrightarrow[\overset{\tau}{[(-1)\mathbf{1}+\mathbf{3}]}]{\overset{\tau}{[(-1)\mathbf{1}+\mathbf{2}]}}\left[\begin{array}{ccc}1&0&0\\1&-1&-1\\1&-1&-1\\ \hline 1&-1&-1\\0&1&0\\0&0&1\end{array}\right]\xrightarrow{\overset{\tau}{[(-1)\mathbf{2}+\mathbf{3}]}}\left[\begin{array}{ccc}1&0&0\\1&-1&0\\1&-1&0\\ \hline 1&-1&\mathbf{0}\\0&1&\mathbf{-1}\\0&0&\mathbf{1}\end{array}\right]$$

$$\left[\begin{array}{c}\mathbf{A}-2\mathbf{I}\\ \hline \mathbf{I}\end{array}\right]=\left[\begin{array}{ccc}-1&1&1\\1&-2&0\\1&0&-2\\ \hline 1&0&0\\0&1&0\\0&0&1\end{array}\right]\xrightarrow[\overset{\tau}{[(1)\mathbf{1}+\mathbf{3}]}]{\overset{\tau}{[(1)\mathbf{1}+\mathbf{2}]}}\left[\begin{array}{ccc}-1&0&0\\1&-1&1\\1&1&-1\\ \hline 1&1&1\\0&1&0\\0&0&1\end{array}\right]\xrightarrow{\overset{\tau}{[(1)\mathbf{2}+\mathbf{3}]}}\left[\begin{array}{ccc}-1&0&0\\1&-1&0\\1&1&0\\ \hline 1&1&\mathbf{2}\\0&1&\mathbf{1}\\0&0&\mathbf{1}\end{array}\right]$$

$$\left[\begin{array}{c}\mathbf{A}+1\mathbf{I}\\ \hline \mathbf{I}\end{array}\right]=\left[\begin{array}{ccc}2&1&1\\1&1&0\\1&0&1\\ \hline 1&0&0\\0&1&0\\0&0&1\end{array}\right]\xrightarrow[\overset{\tau}{[(-1)\mathbf{1}+\mathbf{3}]}]{\overset{\tau}{[(2)\mathbf{2}]}\ \overset{\tau}{[(-1)\mathbf{1}+\mathbf{2}]}\ \overset{\tau}{[(2)\mathbf{3}]}}\left[\begin{array}{ccc}2&0&0\\1&1&-1\\1&-1&1\\ \hline 1&-1&-1\\0&2&0\\0&0&2\end{array}\right]\xrightarrow{\overset{\tau}{[(1)\mathbf{2}+\mathbf{3}]}}\left[\begin{array}{ccc}2&0&0\\1&1&0\\1&-1&0\\ \hline 1&-1&\mathbf{-2}\\0&2&\mathbf{2}\\0&0&\mathbf{2}\end{array}\right]$$

□

(L-18) Question 3. $|\mathbf{A}-\lambda\mathbf{I}|=\lambda\left(9-\lambda^2\right)=0\ \rightarrow\ \begin{cases}\lambda_1=0\\ \lambda_2=3\\ \lambda_3=-3\end{cases}$.

$$\begin{cases}\lambda_1=-3:\ \left[\begin{array}{c}\mathbf{A}+3\mathbf{I}\\ \hline \mathbf{I}\end{array}\right]=\left[\begin{array}{ccc}4&0&2\\0&2&-2\\2&-2&3\\ \hline 1&0&0\\0&1&0\\0&0&1\end{array}\right]\xrightarrow[\overset{\tau}{[(-1)\mathbf{1}+\mathbf{3}]}]{\overset{\tau}{[(2)\mathbf{3}]}}\left[\begin{array}{ccc}4&0&0\\0&2&-4\\2&-2&4\\ \hline 1&0&-1\\0&1&0\\0&0&2\end{array}\right]\xrightarrow{\overset{\tau}{[(2)\mathbf{2}+\mathbf{3}]}}\left[\begin{array}{ccc}4&0&0\\0&2&0\\2&-2&0\\ \hline 1&0&-1\\0&1&2\\0&0&2\end{array}\right]\\ \lambda_1=0:\ \left[\begin{array}{c}\mathbf{A}-0\mathbf{I}\\ \hline \mathbf{I}\end{array}\right]=\left[\begin{array}{ccc}1&0&2\\0&-1&-2\\2&-2&0\\ \hline 1&0&0\\0&1&0\\0&0&1\end{array}\right]\xrightarrow{\overset{\tau}{[(-2)\mathbf{1}+\mathbf{3}]}}\left[\begin{array}{ccc}1&0&0\\0&-1&-2\\2&-2&-4\\ \hline 1&0&-2\\0&1&0\\0&0&1\end{array}\right]\xrightarrow{\overset{\tau}{[(-2)\mathbf{2}+\mathbf{3}]}}\left[\begin{array}{ccc}1&0&0\\0&-1&0\\2&-2&0\\ \hline 1&0&-2\\0&1&-2\\0&0&1\end{array}\right]\\ \lambda_1=3:\ \left[\begin{array}{c}\mathbf{A}-3\mathbf{I}\\ \hline \mathbf{I}\end{array}\right]=\left[\begin{array}{ccc}-2&0&2\\0&-4&-2\\2&-2&-3\\ \hline 1&0&0\\0&1&0\\0&0&1\end{array}\right]\xrightarrow{\overset{\tau}{[(1)\mathbf{1}+\mathbf{3}]}}\left[\begin{array}{ccc}-2&0&0\\0&-4&-2\\2&-2&-1\\ \hline 1&0&1\\0&1&0\\0&0&1\end{array}\right]\xrightarrow[\overset{\tau}{[(-1)\mathbf{2}+\mathbf{3}]}]{\overset{\tau}{[(2)\mathbf{3}]}}\left[\begin{array}{ccc}-2&0&0\\0&-4&0\\2&-2&0\\ \hline 1&0&2\\0&1&-1\\0&0&2\end{array}\right]\end{cases}$$

Thus, for example

$$\text{If}\quad \mathbf{D}=\begin{bmatrix}0&0&0\\0&3&0\\0&0&-3\end{bmatrix}\quad\text{then}\quad \mathbf{Q}=\frac{1}{3}\begin{bmatrix}2&-2&-1\\2&1&2\\-1&-2&2\end{bmatrix}.$$

```
B = Matrix([[1,0,2],[0,-1,-2],[2,-2,0]])
l = sympy.symbols('\lambda')
```

```
d = Determinante( (B-l*I(B.n)) ,1)
p = sympy.poly(d.valor)
e = sympy.real_roots(p)
display(Sistema([d, e]))
S=Sistema([])
D = tuple()
for i in set(e):
    caso = "\lambda = %d\n" % i
    display(Math(caso))
    L = Elim( (B-i*I(B.n)).apila(I(B.n),1), 1)
    cL0 = tuple([c+1 for c,v in enumerate(slice(1,B.n)|L) if v.es_nulo()])
    S = S.concatena(slice(B.n+1,None)|L|cL0).sis()
    D = D+(i,)*e.count(i)

Q = Matrix([v.normalizado() for v in S]).GS()
D = Vector(D).diag()
display(Sistema([D,Q,~Q,Q**-1]))
display(Sistema([(~Q)*B*Q, Q*D*(~Q)]))
~Q*Q
```

O sencillamente

```
B = Matrix([[1,0,2],[0,-1,-2],[2,-2,0]])
espectro = [0,3,-3]
D = DiagonalizaO( B, espectro )
display(Sistema([D, D.Q]))
```

□

(L-18) Question 4(a) $\boldsymbol{u}$, $\boldsymbol{v}$, and $\boldsymbol{w}$ are orthogonal to each other.

□

(L-18) Question 4(b) The null space $\mathcal{N}\left(\mathbf{A}\right)$ is spanned by $\boldsymbol{u}$; that is, it consists of all multiples of $\boldsymbol{u}$; and since the matrix is symmetric, the left null space $\mathcal{N}\left(\mathbf{A}^{\mathsf{T}}\right)$ is identical to $\mathcal{N}\left(\mathbf{A}\right)$. The row space and the column space are also equal to each other ($\mathcal{C}\left(\mathbf{A}\right)$ =$\mathcal{C}\left(\mathbf{A}^{\mathsf{T}}\right)$), and are spanned by all linear combinations of $\boldsymbol{v}$ and $\boldsymbol{w}$.

$$\mathcal{C}\left(\mathbf{A}\right) = \mathcal{C}\left(\mathbf{A}^{\mathsf{T}}\right) \perp \mathcal{N}\left(\mathbf{A}\right) = \mathcal{N}\left(\mathbf{A}^{\mathsf{T}}\right).$$

□

(L-18) Question 4(c) $\boldsymbol{x} = \boldsymbol{v} + \frac{1}{2}\boldsymbol{w}$, since

$$\mathbf{A}(\boldsymbol{v} + \frac{1}{2}\boldsymbol{w}) = \mathbf{A}\boldsymbol{v} + \frac{1}{2}\mathbf{A}\boldsymbol{w} = \boldsymbol{v} + \frac{1}{2}(2\boldsymbol{w}) = \boldsymbol{v} + \boldsymbol{w};$$

but this solution is not unique; we can add any multiple of $\boldsymbol{u}$ (the null space vector) to $\boldsymbol{x}$.

□

(L-18) Question 4(d) Since $\boldsymbol{b} \in \mathcal{C}\left(\mathbf{A}\right) = \mathcal{C}\left(\mathbf{A}^{\mathsf{T}}\right)$ and $\mathcal{N}\left(\mathbf{A}\right)$ is orthogonal to $\mathcal{C}\left(\mathbf{A}^{\mathsf{T}}\right)$ =$\mathcal{C}\left(\mathbf{A}\right)$, we need $\boldsymbol{b} \cdot \boldsymbol{u} = 0$.

□

(L-18) Question 4(e) $\mathbf{S}^{-1} = \mathbf{S}^{\mathsf{T}}$;

$$\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \begin{bmatrix} 0 & & \\ & 1 & \\ & & 2 \end{bmatrix} = \mathbf{D}.$$

□

(L-18) Question 5(a) The eigenvalues are real.

□

(L-18) Question 5(b) All the eigenvalues are less than one in absolute value.

□

(L-18) Question 5(c) It has repeated eigenvalues.

□

(L-18) Question 5(d) It has at least one eigenvalue that is equal to zero.

□

(L-18) Question 6(a) The characteristic equation is $|\mathbf{A} - \lambda\mathbf{I}| = 0$.

$$\begin{vmatrix} -\lambda & 1 & 0 & 0 \\ 0 & -\lambda & 1 & 0 \\ 0 & 0 & -\lambda & 1 \\ 1 & 0 & 0 & -\lambda \end{vmatrix} = -\lambda \begin{vmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 0 & 0 & -\lambda \end{vmatrix} - 1 \begin{vmatrix} 1 & 0 & 0 \\ -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \end{vmatrix} = \lambda^4 - 1 = 0$$

Thus, $\lambda = \pm 1$ and $\lambda = \pm i$; that is, four distinct eigenvalues.

□

(L-18) Question 6(b) The matrix has a null space of dimension 3. Therefore, let us find three eigenvectors that form a basis of the null space of $\mathbf{B}$ (eigenvalue equal to zero):

$$\boldsymbol{x}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}; \quad \boldsymbol{x}_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}; \quad \boldsymbol{x}_3 = \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

In addition,

$$\frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

Therefore, there is a triple eigenvalue equal to 0, and another eigenvalue equal to 1. Since we have found four linearly independent eigenvectors, this matrix is diagonalizable.

□

(L-18) Question 7. If $\mathbf{A}^3 = \mathbf{0}$ then for all eigenvalues $\lambda^3 = 0$, that is, $\lambda = 0$ as in

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

If $\mathbf{A}$ is symmetric, then it is diagonalizable, and by the spectral theorem, it can be factored as

$$\mathbf{A}^3 = \mathbf{Q}\mathbf{D}^3\mathbf{Q}^\intercal = \mathbf{Q}\mathbf{0}\mathbf{Q}^\intercal = \mathbf{0}.$$

□

(L-18) Question 8(a) Los autovalores de

$$\begin{bmatrix} 3 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

son los elementos de la diagonal; $\lambda = 3$ (con multiplicidad 2) y $\lambda = 2$. Pero el rango de

$$\mathbf{A} - 3\lambda = \begin{bmatrix} 3-3 & 1 & 1 \\ 0 & 3-3 & 1 \\ 0 & 0 & 2-3 \end{bmatrix}$$

es 2; por tanto, la matriz no es diagonalizable.

□

(L-18) Question 8(b) Los autovalores de

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

son los elementos de la diagonal; $\lambda = 3$ y $\lambda = 2$ (con multiplicidad 2). El rango de

$$\mathbf{A} - 2\lambda = \begin{bmatrix} 2-2 & 1 & 1 \\ 0 & 3-2 & 1 \\ 0 & 0 & 2-2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

es 1; por tanto, la matriz es diagonalizable.

Dos autovectores independientes correspondientes al autovalor $\lambda = 2$ son $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ y $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$.

Por otra parte

$$\mathbf{A} - 3\lambda = \begin{bmatrix} 2-3 & 1 & 1 \\ 0 & 3-3 & 1 \\ 0 & 0 & 2-3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

Un autovector correspondiente al autovalor $\lambda = 3$ es $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.

Así pues

$$\mathbf{D} = \begin{bmatrix} 2 & & \\ & 2 & \\ & & 3 \end{bmatrix} \quad \text{and} \quad \mathbf{S} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

son matrices tales que $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$.

□

(L-18) Question 8(c) Sea como sea $\mathbf{A}$, la matriz $\mathbf{A}^\intercal\mathbf{A}$ siempre es simétrica; y por tanto es diagonalizable, y es posible encontrar una base ortonormal de autovectores de $\mathbf{A}^\intercal\mathbf{A}$.

□

(L-18) Question 8(d) Basta con encontrar los valores de a que hacen la matriz de rango completo; es decir, cualquier valor de a distinto de cero ($a \neq 0$) (para que la matriz sea invertible) y simultáneamente distinto de tres ($a \neq 3$) (para que la matriz sea diagonalizable).

□

(L-18) Question 9(a) Ecuación característica:

$$0 = |\mathbf{B} - \lambda\mathbf{I}| = \lambda^2 - 1,$$

por tanto $\lambda = \pm 1$

- $\lambda_1 = 1$

$$(\mathbf{B} - \mathbf{I}) = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

Con autovector unitario $\boldsymbol{x}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

- $\lambda_2 = -1$

$$(\mathbf{B} + \mathbf{I}) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Con autovector unitario $\boldsymbol{x}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \end{pmatrix}$

Por tanto

$$\mathbf{B} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\intercal = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & \\ & -1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\frac{1}{\sqrt{2}}$$

□

(L-18) Question 9(b) Puesto que los dos autovalores son distintos, los dos autovectores encontrados son linealmente independientes, y la matriz es diagonalizable. Una matriz $\mathbf{S}$ que diagonaliza $\mathbf{B}$ es

$$\mathbf{S} = \begin{bmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix};$$

ya que $\mathbf{BS} = \mathbf{SD}$, donde

$$\mathbf{D} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Puesto que la matriz $\mathbf{S}$ es invertible, y dado que ambos autovectores son linealmente independientes, podemos diagonalizar $\mathbf{B}$ del siguiente modo:

$$\mathbf{S}^{-1}\mathbf{BS} = \mathbf{D};$$

donde la matriz diagonal contiene los autovalores de $\mathbf{B}$.

□

(L-18) Question 10(a) The three equations tell us that $\mathbf{AQ} = \mathbf{I}$, where the columns of $\mathbf{Q}$ are $\boldsymbol{q}_1$, $\boldsymbol{q}_2$, $\boldsymbol{q}_3$. So $\mathbf{A} = \mathbf{Q}^{-1}$, and since $\mathbf{Q}$ is an orthogonal matrix, we have $\mathbf{A} = \mathbf{Q}^{-1} = \mathbf{Q}^{\intercal} = \begin{bmatrix}\boldsymbol{q}_1; & \boldsymbol{q}_2; & \boldsymbol{q}_3;\end{bmatrix}^{\intercal}$.

□

(L-18) Question 10(b) ComoSince $\mathbf{A} = \mathbf{Q}^{\intercal}$ then $\det \mathbf{A} = \det \mathbf{Q}$; and since $\mathbf{Q}^{\intercal}\mathbf{Q} = \mathbf{I}$

$$1 = \det \mathbf{I} = \det\left(\mathbf{Q}^{\intercal}\mathbf{Q}\right) = \det\left(\mathbf{Q}^{\intercal}\right)\det\left(\mathbf{Q}\right) = |\mathbf{Q}|^2 \quad \Rightarrow \quad \det \mathbf{A} \pm 1.$$

□

(L-18) Question 10(c) The eigenvalues of $\mathbf{A}$ must have absolute value $|\lambda| = 1$.

Proof: Let λ be an eigenvalue of $\mathbf{A}$ and let $\boldsymbol{v}$ be a corresponding eigenvector. Then we have

$$\mathbf{A}\boldsymbol{v} = \lambda\boldsymbol{v}.$$

It follows from this that

$$\|\mathbf{A}\boldsymbol{v}\|^2 = \|\lambda\boldsymbol{v}\|^2 = |\lambda|^2\|\boldsymbol{v}\|^2.$$

And since $\mathbf{A}$ is orthogonal ($\mathbf{A}^{\intercal}\mathbf{A} = \mathbf{I}$), the left-hand side becomes

$$\|\mathbf{A}\boldsymbol{v}\|^2 \;=\; \mathbf{A}\boldsymbol{v}\cdot\mathbf{A}\boldsymbol{v} \;=\; \boldsymbol{v}\mathbf{A}^{\intercal}\mathbf{A}\boldsymbol{v} \;=\; \boldsymbol{v}\cdot\boldsymbol{v} \;=\; \|\boldsymbol{v}\|^2$$

It follows that

$$|\lambda|^2\|\boldsymbol{v}\|^2 = \|\boldsymbol{v}\|^2.$$

Since $\boldsymbol{v}$ is an eigenvector, it is non-zero, and hence $\|\boldsymbol{v}\| \neq \mathbf{0}$. Canceling $\|\boldsymbol{v}\|$, we have $|\lambda|^2 = 1$. Since the absolute value is non-negative, we obtain $\boxed{|\lambda| = 1}$.

□

(L-19) Question 1(a) No. $\begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix} \xrightarrow[[(-3)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \begin{bmatrix} 1 & 0 \\ 3 & -4 \end{bmatrix} \xrightarrow[[(-3)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \begin{bmatrix} 1 & 0 \\ 0 & -4 \end{bmatrix}$. $\quad f(x,y) = x^2 + 6xy + 5y^2$.

```
A = Matrix([[1,3],[3,5]])
DiagonalizaC(A,1)
```

□

(L-19) Question 1(b) No. $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \xrightarrow[[(1)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} \xrightarrow[[(1)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$. $\quad f(x,y) = x^2 - 2xy + y^2 = (x-y)^2$.

□

(L-19) Question 1(c) Yes. $\begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} \xrightarrow[[(-3)\mathbf{1}+\mathbf{2}]]{\substack{\boldsymbol{\tau} \\ [(2)\mathbf{2}]}} \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix} \xrightarrow[\substack{[(-3)\mathbf{1}+\mathbf{2}] \\ [(2)\mathbf{2}]}]{\boldsymbol{\tau}} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$. $\quad f(x,y) = 2x^2 + 6xy + 5y^2$.

□

(L-19) Question 1(d) No, There are negative components in the diagonal. $\begin{bmatrix} -1 & 2 \\ 2 & -8 \end{bmatrix} \xrightarrow[[(2)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \begin{bmatrix} -1 & 0 \\ 2 & -4 \end{bmatrix} \xrightarrow[[(2)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}}$

$\begin{bmatrix} -1 & 0 \\ 0 & -4 \end{bmatrix}$. $\quad f(x,y) = -x^2 + 4xy - 8y^2$.

□

(L-19) Question 1(e) The function $f(x,y)=(y-x)^2$ is equal to zero along the line $y=x$. Note that (1,1) is an eigenvector corresponding to $\lambda=0$.

□

(L-19) Question 2(a) $\begin{bmatrix}1&2\\2&9\end{bmatrix} \xrightarrow[[(-2)\mathbf{1}+\mathbf{2}]]{\tau} \begin{bmatrix}1&0\\2&5\end{bmatrix} \xrightarrow[[(-2)\mathbf{1}+\mathbf{2}]]{\tau} \begin{bmatrix}1&0\\0&5\end{bmatrix}$. That is

$$\mathbf{D}=\begin{bmatrix}1&0\\0&5\end{bmatrix}=\begin{bmatrix}1&0\\-2&1\end{bmatrix}\begin{bmatrix}1&2\\2&9\end{bmatrix}\begin{bmatrix}1&-2\\0&1\end{bmatrix}=\mathbf{B}^{\intercal}\mathbf{A}\mathbf{B}.$$

Therefore, $\boldsymbol{x}\mathbf{A}\boldsymbol{x}=\boldsymbol{x}\left(\mathbf{B}^{-1}\right)^{\intercal}\mathbf{D}\mathbf{B}^{-1}\boldsymbol{x}=\begin{pmatrix}x&y\end{pmatrix}\begin{bmatrix}1&0\\2&1\end{bmatrix}\begin{bmatrix}1&0\\0&5\end{bmatrix}\begin{bmatrix}1&2\\0&1\end{bmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}x+2y&y\end{pmatrix}\begin{bmatrix}1&0\\0&5\end{bmatrix}\begin{pmatrix}x+2y\\y\end{pmatrix}$.

Thus, $f(x,y)=x^2+4\,x\,y+9\,y^2=(x+2y)^2+5(y)^2$.

□

(L-19) Question 2(b) $\begin{bmatrix}1&3\\3&9\end{bmatrix} \xrightarrow[[(-3)\mathbf{1}+\mathbf{2}]]{\tau} \begin{bmatrix}1&0\\3&0\end{bmatrix} \xrightarrow[[(-3)\mathbf{1}+\mathbf{2}]]{\tau} \begin{bmatrix}1&0\\0&0\end{bmatrix}$. $\quad f(x,y)=x^2+6\,x\,y+9\,y^2=(x+3y)^2+0(y)^2$.

□

(L-19) Question 3(a) Not this one, since the determinant of $\mathbf{A}$ is negative.

□

(L-19) Question 3(b) Not this one; since $a=-1$.

□

(L-19) Question 3(c) Not this one; this matrix is singular ($\det\mathbf{C}=0$).

□

(L-19) Question 3(d) $\mathbf{D}$ has two positive eigenvalues since $a=1$ and $\det\mathbf{A}=1$.

□

(L-19) Question 4. $\begin{bmatrix}1&2\\2&3\end{bmatrix} \xrightarrow[[(-2)\mathbf{1}+\mathbf{2}]]{\tau} \begin{bmatrix}1&0\\2&-1\end{bmatrix} \xrightarrow[[(-2)\mathbf{1}+\mathbf{2}]]{\tau} \begin{bmatrix}1&0\\0&-1\end{bmatrix}$.

$$f(x,y)=x^2+4xy+3y^2=(x+2y)^2-y^2.$$

It is negative (because $-y^2$) at $(2,-1)$:

$$f(2,-1)=-1.$$

□

(L-19) Question 5. If $\mathbf{A}\boldsymbol{x}=\lambda\boldsymbol{x} \quad\Rightarrow\quad \mathbf{A}^2\boldsymbol{x}=\mathbf{A}\mathbf{A}\boldsymbol{x}=\mathbf{A}(\lambda\boldsymbol{x})=\lambda^2\boldsymbol{x}$.

Si $\mathbf{A}$ tiene autovalores λ_i positivos, entonces los autovalores de $\mathbf{A}^2$ son λ_i^2 y los de $\mathbf{A}^{-1}$ son $1/\lambda_i$, y por tanto también positivos.

□

(L-19) Question 6(a) The corresponding matrix is

$$\begin{bmatrix}1&-2&0\\-2&4&0\\0&0&5\end{bmatrix} \xrightarrow[[(2)\mathbf{1}+\mathbf{2}]]{\tau} \begin{bmatrix}1&0&0\\0&0&0\\0&0&5\end{bmatrix} \xrightarrow[[\mathbf{2}\rightleftharpoons\mathbf{3}]]{\tau} \begin{bmatrix}1&0&0\\0&5&0\\0&0&0\end{bmatrix}$$

Cuya diagonal está formada por dos números positivos y un cero.

```
A = Matrix([[1,-2,0],[-2,4,0],[0,0,5]])
DiagonalizaC(A,1)
```

□

(L-19) Question 6(b) The corresponding matrix is

$$\mathbf{A} = \begin{bmatrix} -1 & 1 & -a \\ 1 & 4 & 0 \\ -a & 0 & 1 \end{bmatrix} \xrightarrow[\substack{[(1)\mathbf{1}+\mathbf{2}] \\ [(-a)\mathbf{1}+\mathbf{3}]}]{\boldsymbol{\tau}} \begin{bmatrix} -1 & 0 & 0 \\ 1 & 5 & -a \\ -a & -a & a^2+1 \end{bmatrix} \xrightarrow[\substack{[(-a)\mathbf{1}+\mathbf{3}] \\ [(1)\mathbf{1}+\mathbf{2}]}]{\boldsymbol{\tau}} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 5 & -a \\ 0 & -a & a^2+1 \end{bmatrix} \dots$$

Since there is a 5 (positive) in the main diagonal, it cannot be negative definite.

□

(L-19) Question 7(a) $\mathbf{A}$ is not positive definite since $\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^\intercal$ is in its nullspace. In fact, diagonalizing by congruence we see it is positive semidefinite

$$\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \xrightarrow[\substack{[(2)\mathbf{2}] \\ [(1)\mathbf{1}+\mathbf{2}] \\ [(2)\mathbf{3}] \\ [(1)\mathbf{1}+\mathbf{3}]}]{\boldsymbol{\tau}} \begin{bmatrix} 2 & 0 & 0 \\ -1 & 3 & -3 \\ -1 & -3 & 3 \end{bmatrix} \xrightarrow[\substack{[(1)\mathbf{1}+\mathbf{3}] \\ [(2)\mathbf{3}] \\ [(1)\mathbf{1}+\mathbf{2}] \\ [(2)\mathbf{2}]}]{\boldsymbol{\tau}} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & -6 \\ 0 & -6 & 6 \end{bmatrix} \xrightarrow[[(1)\mathbf{2}+\mathbf{3}]]{\boldsymbol{\tau}} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & -6 & 0 \end{bmatrix} \xrightarrow[[(1)\mathbf{2}+\mathbf{3}]]{\boldsymbol{\tau}} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

□

(L-19) Question 7(b) $\mathbf{B}$ is positive definite

$$\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix} \xrightarrow[\substack{[(2)\mathbf{2}] \\ [(1)\mathbf{1}+\mathbf{2}] \\ [(2)\mathbf{3}] \\ [(1)\mathbf{1}+\mathbf{3}]}]{\boldsymbol{\tau}} \begin{bmatrix} 2 & 0 & 0 \\ -1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix} \xrightarrow[\substack{[(1)\mathbf{1}+\mathbf{3}] \\ [(2)\mathbf{3}] \\ [(1)\mathbf{1}+\mathbf{2}] \\ [(2)\mathbf{2}]}]{\boldsymbol{\tau}} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 2 \\ 0 & 2 & 6 \end{bmatrix} \xrightarrow[[(-1)\mathbf{2}+\mathbf{3}]]{\boldsymbol{\tau}} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 2 & 16 \end{bmatrix} \xrightarrow[\substack{[(-1)\mathbf{2}+\mathbf{3}] \\ [(3)\mathbf{3}]}]{\boldsymbol{\tau}} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 48 \end{bmatrix}$$

Hence, Sylvester's method works in this case!... $\det(2) = 2$; $\begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 3$; $\det(\mathbf{B}) = 4$.

□

(L-19) Question 7(c) The eigenvalues of $\mathbf{C}$ are the squares of the eigenvalues of

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix};$$

therefore, if this matrix is full rank, then $\mathbf{C}$ is positive definite. Let's see it through congruent diagonalization:

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \xrightarrow[[(1)\mathbf{2}+\mathbf{1}]]{\boldsymbol{\tau}} \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 3 & 1 & 0 \end{bmatrix} \xrightarrow[[(1)\mathbf{2}+\mathbf{1}]]{\boldsymbol{\tau}} \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 1 \\ 3 & 1 & 0 \end{bmatrix} \xrightarrow[\substack{[(2)\mathbf{2}] \\ [(-1)\mathbf{1}+\mathbf{2}] \\ [(2)\mathbf{3}] \\ [(-3)\mathbf{1}+\mathbf{3}]}]{\boldsymbol{\tau}} \begin{bmatrix} 2 & 0 & 0 \\ 1 & -1 & -1 \\ 3 & -1 & -9 \end{bmatrix}$$

$$\xrightarrow[\substack{[(-3)\mathbf{1}+\mathbf{3}] \\ [(2)\mathbf{3}] \\ [(-1)\mathbf{1}+\mathbf{2}] \\ [(2)\mathbf{2}]}]{\boldsymbol{\tau}} \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & -2 \\ 0 & -2 & -18 \end{bmatrix} \xrightarrow[[(-1)\mathbf{2}+\mathbf{3}]]{\boldsymbol{\tau}} \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & -2 & -16 \end{bmatrix} \xrightarrow[[(-1)\mathbf{2}+\mathbf{3}]]{\boldsymbol{\tau}} \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -16 \end{bmatrix}.$$

Since there are three pivots, zero is not an eigenvalue of that matrix nor of $\mathbf{C}$. Therefore, $\mathbf{C}$ is positive definite.

□

(L-19) Question 8.

$$\begin{bmatrix}1&3&0\\3&1&0\\0&0&a\end{bmatrix} \xrightarrow[[(-3)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \begin{bmatrix}1&0&0\\3&-8&0\\0&0&a\end{bmatrix} \xrightarrow[[(-3)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \begin{bmatrix}1&0&0\\0&-8&0\\0&0&a\end{bmatrix};$$

la forma cuadrática no es definida (cualquiera que sea el valor de a).

□

(L-19) Question 9. $\mathbf{A}$ is symmetric and therefore diagonalizable:

$$\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1};$$

and we know that in this case

$$\mathbf{A}^n = \mathbf{S}\mathbf{D}^n\mathbf{S}^{-1};$$

Thus, since $\mathbf{A}$ is positive definite, its eigenvalues are positive $\lambda_1 > 0$ and $\lambda_2 > 0$, and therefore $\lambda_1^{-1} > 0$ and $\lambda_2^{-1} > 0$, which are the eigenvalues of $\mathbf{A}^{-1}$.

□

(L-19) Question 10. The characteristic equation is:

$$|\mathbf{A} - \lambda\mathbf{I}| = \begin{vmatrix} a-\lambda & b \\ b & c-\lambda \end{vmatrix} = \lambda^2 - (a+c)\lambda + ac - b^2 = 0;$$

whose solutions are

$$\lambda = \frac{(a+c) \pm \sqrt{(a+c)^2 - 4(ac-b^2)}}{2}$$

but

$$(a+c)^2 - 4(ac-b^2) = a^2 + c^2 + 2ac - 4ac + 4b^2 = a^2 + c^2 - 2ac + 4b^2 = (a-c)^2 + (2b)^2$$

which is a sum of squares, and therefore, what is inside the square root is greater than or equal to zero. Thus, the eigenvalues are real (there are no square roots of negative numbers).

On the other hand, if $a > 0$ and $ac > b^2$, then $c \geq 0$. So we know that $(a+c) > 0$ and therefore

$$\lambda_1 = \frac{(a+c) + \sqrt{(a-c)^2 + (2b)^2}}{2} > 0$$

Furthermore, since $ac > b^2$, we know that $\det \mathbf{A} = ac - b^2 > 0$; but since $\lambda_1\lambda_2 = \det \mathbf{A} > 0$ and $\lambda_1 > 0$, it follows that $\lambda_2 > 0$.

□

(L-19) Question 11(a) Let's examine the signs of the pivots.

$$\begin{bmatrix}1&2&3\\2&5&4\\3&4&9\end{bmatrix} \xrightarrow[\substack{[(-2)\mathbf{1}+\mathbf{2}]\\[(-3)\mathbf{1}+\mathbf{3}]}]{\boldsymbol{\tau}} \begin{bmatrix}1&0&0\\2&1&-2\\3&-2&0\end{bmatrix} \xrightarrow[\substack{[(-3)\mathbf{1}+\mathbf{3}]\\[(-2)\mathbf{1}+\mathbf{2}]}]{\boldsymbol{\tau}} \begin{bmatrix}1&0&0\\0&1&-2\\0&-2&0\end{bmatrix} \xrightarrow[[(2)\mathbf{2}+\mathbf{3}]]{\boldsymbol{\tau}} \begin{bmatrix}1&0&0\\0&1&0\\0&-2&-4\end{bmatrix} \xrightarrow[[(2)\mathbf{2}+\mathbf{3}]]{\boldsymbol{\tau}} \begin{bmatrix}1&0&0\\0&1&0\\0&0&-4\end{bmatrix}$$

Since there are both positive and negative pivots, the matrix is **indefinite**.

□

(L-19) Question 11(b)

$$\begin{bmatrix}1&2&0&0\\2&6&-2&0\\0&-2&5&-2\\0&0&-2&3\end{bmatrix} \xrightarrow[[(-2)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \begin{bmatrix}1&0&0&0\\2&2&-2&0\\0&-2&5&-2\\0&0&-2&3\end{bmatrix} \xrightarrow[[(1)\mathbf{2}+\mathbf{3}]]{\boldsymbol{\tau}} \begin{bmatrix}1&0&0&0\\2&2&0&0\\0&-2&3&-2\\0&0&-2&3\end{bmatrix} \xrightarrow[\left[\left(\frac{2}{3}\right)\mathbf{3}+\mathbf{4}\right]]{\boldsymbol{\tau}} \begin{bmatrix}1&0&0&0\\2&2&0&0\\0&-2&3&0\\0&0&-2&\frac{5}{3}\end{bmatrix}$$

Since by applying elimination using only Type I transformations we reached a triangular matrix whose pivots are positive → **Definida positiva.**

□

(L-19) Question 11(c) Since $\mathbf{B}$ is positive definite, $-\mathbf{B}$ is **Negative Definite**.

□

(L-19) Question 11(d) Since $\mathbf{A}$ has two positive pivots $\lambda_1 > 0$ and $\lambda_2 > 0$ and one negative $\lambda_3 < 0$; the pivots of $\mathbf{D}$ —which are the inverses of those of $\mathbf{A}$— maintain their signs. Hence, it is: **indefinite**.
(exercise 14 of the problem set 6.2 from the textbook)

□

(L-19) Question 12. $\begin{pmatrix} 0 & a & 0 \end{pmatrix}$, with $a \neq 0$

□

(L-19) Question 13. $\boldsymbol{x}(\mathbf{A}+\mathbf{B})\boldsymbol{x} = (\boldsymbol{x}\mathbf{A}+\boldsymbol{x}\mathbf{B})\boldsymbol{x} = \boldsymbol{x}\mathbf{A}\boldsymbol{x}+\boldsymbol{x}\mathbf{B}\boldsymbol{x} > 0$, since $\mathbf{A}$ and $\mathbf{B}$ are positive definite.

□

(L-19) Question 14(a)

$$\left[\begin{array}{c}\mathbf{A}\\\hline \mathbf{I}\end{array}\right] = \left[\begin{array}{cccc} 1&1&1&1\\ 1&2&3&4\\ 1&3&6&10\\ 1&4&10&20\\ \hline 1&0&0&0\\ 0&1&0&0\\ 0&0&1&0\\ 0&0&0&1\end{array}\right] \xrightarrow[\substack{[(-1)\mathbf{1}+\mathbf{2}]\\[(-1)\mathbf{1}+\mathbf{3}]\\[(-1)\mathbf{1}+\mathbf{4}]}]{\boldsymbol{\tau}} \left[\begin{array}{cccc} 1&0&0&0\\ 1&1&2&3\\ 1&2&5&9\\ 1&3&9&19\\ \hline 1&-1&-1&-1\\ 0&1&0&0\\ 0&0&1&0\\ 0&0&0&1\end{array}\right] \xrightarrow[\substack{[(-2)\mathbf{2}+\mathbf{3}]\\[(-3)\mathbf{2}+\mathbf{4}]}]{\boldsymbol{\tau}}$$

$$\left[\begin{array}{cccc} 1&0&0&0\\ 1&1&0&0\\ 1&2&1&3\\ 1&3&3&10\\ \hline 1&-1&1&2\\ 0&1&-2&-3\\ 0&0&1&0\\ 0&0&0&1\end{array}\right] \xrightarrow[[(-3)\mathbf{3}+\mathbf{4}]]{\boldsymbol{\tau}} \left[\begin{array}{cccc} 1&0&0&0\\ 1&1&0&0\\ 1&2&1&0\\ 1&3&3&1\\ \hline 1&-1&1&-1\\ 0&1&-2&3\\ 0&0&1&-3\\ 0&0&0&1\end{array}\right] = \left[\begin{array}{c}\mathbf{L}\\\hline \dot{\mathbf{E}}\end{array}\right]$$

Hence, the Gauss transformations are

$$\mathbf{G}_{1\triangleright} = \begin{bmatrix} 1&-1&-1&-1\\ 0&1&0&0\\ 0&0&1&0\\ 0&0&0&1\end{bmatrix}; \quad \mathbf{G}_{2\triangleright} = \begin{bmatrix} 1&0&0&0\\ 0&1&-2&-3\\ 0&0&1&0\\ 0&0&0&1\end{bmatrix}; \quad \mathbf{G}_{3\triangleright} = \begin{bmatrix} 1&0&0&0\\ 0&1&0&0\\ 0&0&1&-3\\ 0&0&0&1\end{bmatrix};$$

$\dot{\mathbf{E}} = \dot{\mathbf{G}}_{1\triangleright}\ \dot{\mathbf{G}}_{2\triangleright}\ \dot{\mathbf{G}}_{3\triangleright}$. Matrix $\dot{\mathbf{U}}$ is the inverse of $\dot{\mathbf{E}}^{-1}$, so: $\dot{\mathbf{U}} = \dot{\mathbf{E}}^{-1} = (\dot{\mathbf{G}}_{1\triangleright}\dot{\mathbf{G}}_{2\triangleright}\dot{\mathbf{G}}_{3\triangleright})^{-1} = \dot{\mathbf{G}}_{3\triangleright}^{-1}\dot{\mathbf{G}}_{2\triangleright}^{-1}\dot{\mathbf{G}}_{1\triangleright}^{-1}$, and the $\dot{\mathbf{L}}\mathbf{D}\dot{\mathbf{U}}$ factorization is:

$$\mathbf{A} = \begin{bmatrix} 1&1&1&1\\ 1&2&3&4\\ 1&3&6&10\\ 1&4&10&20\end{bmatrix} = \begin{bmatrix} 1&&&\\ 1&1&&\\ 1&2&1&\\ 1&3&3&1\end{bmatrix}\begin{bmatrix} 1&&&\\ &1&&\\ &&1&\\ &&&1\end{bmatrix}\begin{bmatrix} 1&1&1&1\\ &1&2&3\\ &&1&3\\ &&&1\end{bmatrix} = \dot{\mathbf{L}}\mathbf{D}\dot{\mathbf{U}}$$

that is equal to $\mathbf{A} = \dot{\mathbf{L}}\mathbf{D}\dot{\mathbf{L}}^{\mathsf{T}}$, since $\mathbf{A}$ is symmetric.

□

(L-19) Question 14(b)

$$\left[\begin{array}{c}\mathbf{A}\\\hline \mathbf{I}\end{array}\right] = \left[\begin{array}{cccc} 1&1&1&1\\ 1&2&2&2\\ 1&2&3&3\\ 1&2&3&4\\ \hline 1&0&0&0\\ 0&1&0&0\\ 0&0&1&0\\ 0&0&0&1\end{array}\right] \xrightarrow[\substack{[(-1)\mathbf{1}+\mathbf{2}]\\[(-1)\mathbf{1}+\mathbf{3}]\\[(-1)\mathbf{1}+\mathbf{4}]}]{\boldsymbol{\tau}} \left[\begin{array}{cccc} 1&0&0&0\\ 1&1&1&1\\ 1&1&2&2\\ 1&1&2&3\\ \hline 1&-1&-1&-1\\ 0&1&0&0\\ 0&0&1&0\\ 0&0&0&1\end{array}\right] \xrightarrow[\substack{[(-1)\mathbf{2}+\mathbf{3}]\\[(-1)\mathbf{2}+\mathbf{4}]}]{\boldsymbol{\tau}}$$

$$\left[\begin{array}{cccc} 1&0&0&0\\ 1&1&0&0\\ 1&1&1&1\\ 1&1&1&2\\ \hline 1&-1&0&0\\ 0&1&-1&-1\\ 0&0&1&0\\ 0&0&0&1\end{array}\right] \xrightarrow[[(-1)\mathbf{3}+\mathbf{4}]]{\boldsymbol{\tau}} \left[\begin{array}{cccc} 1&0&0&0\\ 1&1&0&0\\ 1&1&1&0\\ 1&1&1&1\\ \hline 1&-1&0&0\\ 0&1&-1&0\\ 0&0&1&-1\\ 0&0&0&1\end{array}\right] = \left[\begin{array}{c}\mathbf{L}\\\hline \dot{\mathbf{E}}\end{array}\right]$$

Hence, the Gauss transformations are

$$\dot{\mathbf{G}}_{1\triangleright} = \begin{bmatrix}1&-1&-1&-1\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix}; \quad \dot{\mathbf{G}}_{2\triangleright} = \begin{bmatrix}1&0&0&0\\0&1&-1&-1\\0&0&1&0\\0&0&0&1\end{bmatrix}; \quad \dot{\mathbf{G}}_{3\triangleright} = \begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&-1\\0&0&0&1\end{bmatrix};$$

$\dot{\mathbf{E}} = \dot{\mathbf{G}}_{1\triangleright}\,\dot{\mathbf{G}}_{2\triangleright}\,\dot{\mathbf{G}}_{3\triangleright}$. Matrix $\dot{\mathbf{U}}$ is the inverse of $\dot{\mathbf{E}}^{-1}$, so: $\dot{\mathbf{U}} = \dot{\mathbf{E}}^{-1} = \dot{\mathbf{G}}_{3\triangleright}^{-1}\dot{\mathbf{G}}_{2\triangleright}^{-1}\dot{\mathbf{G}}_{1\triangleright}^{-1}$, and the $\dot{\mathbf{L}}\mathbf{D}\dot{\mathbf{U}}$ factorization is:

$$\mathbf{A} = \begin{bmatrix}1&1&1&1\\1&2&2&2\\1&2&3&3\\1&2&3&4\end{bmatrix} = \begin{bmatrix}1&&&\\1&1&&\\1&1&1&\\1&1&1&1\end{bmatrix}\begin{bmatrix}1&&&\\&1&&\\&&1&\\&&&1\end{bmatrix}\begin{bmatrix}1&1&1&1\\&1&1&1\\&&1&1\\&&&1\end{bmatrix} = \dot{\mathbf{L}}\mathbf{D}\dot{\mathbf{U}}$$

that is equal to $\mathbf{A} = \dot{\mathbf{L}}\mathbf{D}\dot{\mathbf{L}}^{\mathsf{T}}$, since $\mathbf{A}$ is symmetric.

□

(L-19) Question 15.

$$\begin{bmatrix}3&6\\6&16\end{bmatrix} \xrightarrow[[(-2)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \begin{bmatrix}3&0\\6&4\end{bmatrix} \xrightarrow[[(-2)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \begin{bmatrix}3&0\\0&4\end{bmatrix}$$

Therefore,

$$\mathbf{A} = \begin{bmatrix}3&6\\6&16\end{bmatrix} = \begin{bmatrix}1&0\\2&1\end{bmatrix}\begin{bmatrix}3&0\\0&4\end{bmatrix}\begin{bmatrix}1&2\\0&1\end{bmatrix}$$

The coefficients of the squares are the pivots in $\mathbf{D}$, and the coefficients within the squares are the columns of $\mathbf{L}$.

□

(L-19) Question 16(a)

$$|\mathbf{A}-\lambda\mathbf{I}| = \begin{vmatrix}2-\lambda&1&0&0\\1&2-\lambda&0&0\\0&0&a-\lambda&0\\0&0&a&a-\lambda\end{vmatrix} = (a-\lambda)^2\begin{vmatrix}2-\lambda&1\\1&2-\lambda\end{vmatrix} = 0.$$

Therefore, the eigenvalue $\lambda = a$ is repeated twice. We can find the other two eigenvalues by solving

$$\begin{vmatrix}2-\lambda&1\\1&2-\lambda\end{vmatrix} = (2-\lambda)^2 - 1 = 0; \qquad \Rightarrow \lambda^2 - 4\lambda + 3 = 0$$

Thus, the other two eigenvalues are 1 and 3.

□

(L-19) Question 16(b) When $\lambda = a = 2$, the matrix

$$\mathbf{A} - 2\mathbf{I} = \begin{bmatrix} 2-\lambda & 1 & 0 & 0 \\ 1 & 2-\lambda & 0 & 0 \\ 0 & 0 & 2-\lambda & 0 \\ 0 & 0 & 2 & 2-\lambda \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \end{bmatrix}$$

has rank 3 (it can be seen immediately that there are three pivot columns). Thus, in this case, the solution set of the system $\mathbf{A}\boldsymbol{x} = \mathbf{0}$ has dimension 1 (four columns minus the rank); therefore, it is NOT possible to find two linearly independent eigenvectors for the repeated eigenvalue $\lambda = 2$, and consequently the matrix is NOT DIAGONALIZABLE.

□

(L-19) Question 16(c)

$$|\mathbf{B} - \lambda\mathbf{I}| = \begin{vmatrix} 2-\lambda & 1 & 0 \\ 1 & 2-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{vmatrix} = (1-\lambda)\cdot\Big((2-\lambda)^2 - 1\Big) = 0$$

Clearly one eigenvalue is $\lambda = 1$. The other two are the roots of

$$\Big((2-\lambda)^2 - 1\Big) = 4 + \lambda^2 - 4\lambda - 1 = \lambda^2 - 4\lambda + 3 = 0.$$

that is, $\lambda = 3$ and $\lambda = 1$. Thus, $\boxed{\mathbf{D} = \left[\begin{smallmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right]}$.

- For $\lambda = 3$

$$\mathbf{A} - 3\lambda\mathbf{I} = \begin{bmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix};$$

therefore $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ is an eigenvector corresponding to $\lambda = 3$. Since its norm is $\sqrt{2}$, the vector $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ is a normalized eigenvector for $\lambda = 3$.

- For $\lambda = 1$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix};$$

thus, the last column of $(\mathbf{A} - \lambda\mathbf{I})$, the vector $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ is an eigenvector (with norm 1). Moreover, from the other two columns, it is easy to check that $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$ is another eigenvector for $\lambda = 3$ with norm $\sqrt{2}$, thus a second normalized eigenvector is $\frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$.

It is straightforward to verify that these three eigenvectors with norm one are orthogonal to each other. Therefore, one possible orthonormal matrix $\mathbf{P}$ is

$$\boxed{\mathbf{P} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}}.$$

□

(L-19) Question 16(d) The quadratic form is

$$f(x,y,z) = \boldsymbol{b}\mathbf{X}\boldsymbol{b} = \begin{pmatrix} x & y & z \end{pmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 2x^2 + 2y^2 + z^2 + 2xy,$$

and we already know it is positive definite, since we have seen that the three eigenvalues of $\mathbf{B}$ are positive (3, 1, and 1).

□

(L-19) Question 17(a) There are two possibilities:

- $a = -4/5$ and $b = 3/5$
- $a = 4/5$ and $b = -3/5$.

□

(L-19) Question 17(b) Any values of a and b that form a vector that is not a multiple of the second column of $\mathbf{A}$; for example, $a = 1$ and $b = 0$.

□

(L-19) Question 17(c) This is the opposite case; we need the matrix to be singular, therefore we need the first column to be a multiple of the second; for example: $a = 3$ and $b = 4$.

□

(L-19) Question 17(d) Since it is symmetric, we must have $b = 3/5$. Additionally, we require that $a < 0$ and that $\det \mathbf{A} > 0$; that is $a \cdot 4/5 - (3/5)^2 > 0$, or

$$a \cdot 4/5 > (3/5)^2$$

which is not possible if a must also be negative. Therefore, THERE ARE NO SUCH VALUES OF a AND b.

□

(L-19) Question 18(a)

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & a & 0 \\ 0 & 0 & 8 \end{bmatrix} \xrightarrow[[(-1)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \begin{bmatrix} 1 & 0 & 0 \\ 1 & a-1 & 0 \\ 0 & 0 & 8 \end{bmatrix} \xrightarrow[[(-1)\mathbf{1}+\mathbf{2}]]{\boldsymbol{\tau}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & a-1 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

Si $a = 1$ la matriz $\mathbf{Q}$ es semidefinida positiva.

□

(L-19) Question 18(b)

- Si $a > 1$ la matriz $\mathbf{Q}$ es definida positiva.
- Si $a < 1$ la matriz $\mathbf{Q}$ es indefinida.

□

(L-Opt-2) Question 1(a)

$$\det \mathbf{A} = \begin{vmatrix} 1 & 1 & 0 & 1 \\ 1 & a & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 & 1 \\ 1 & a & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix} = - \begin{vmatrix} 1 & 0 & 0 & 0 \\ 1 & a & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{vmatrix} = -a;$$

therefore, the determinant $|\mathbf{A}|$ is non-zero if and only if $a \neq 0$.

□

(L-Opt-2) Question 1(b) No, since:

$$|1| = 1; \quad \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0; \quad \begin{vmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 0; \quad \begin{vmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix} = -1,$$

when all determinants should be positive. Thus, the matrix is not definite.

□

(L-Opt-2) Question 1(c)

$$\left[\begin{array}{cccc} 1 & 1 & 0 & 1 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ \hline 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[(-1)\mathbf{1}+\mathbf{2}] \\ [(-1)\mathbf{1}+\mathbf{4}] \\ [(1)\mathbf{2}+\mathbf{4}]}]{\tau} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & -1 & 0 & -2 \\ \hline 1 & -1 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[(-1)\mathbf{2}+\mathbf{1}] \\ [(1)\mathbf{4}+\mathbf{1}] \\ [(2)\mathbf{2}] \\ [(-1)\mathbf{4}+\mathbf{2}]}]{\tau} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -2 \\ \hline 0 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{array}\right] \xrightarrow[\substack{\left[\left(\frac{1}{2}\right)\mathbf{2}\right] \\ \left[\left(\frac{-1}{2}\right)\mathbf{4}\right]}]{\tau} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \hline 0 & 0 & 0 & 1 \\ 0 & \frac{1}{2} & 0 & \frac{-1}{2} \\ 0 & 0 & 1 & 0 \\ 1 & \frac{-1}{2} & 0 & \frac{-1}{2} \end{array}\right]$$

Thus,

$$\mathbf{A}^{-1} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1/2 & 0 & -1/2 \\ 0 & 0 & 1 & 0 \\ 1 & -1/2 & 0 & -1/2 \end{bmatrix}$$

□

(L-Opt-2) Question 1(d) From the steps provided in the first part, it is easy to see that when $a = 0$, the rank of $\mathbf{A}$ is three; therefore, there are three endogenous variables (or pivot variables). Thus, only one variable can be exogenous (or free).

Since when $a = 0$ the second and fourth columns are equal (and hence dependent), we can take either the second or the fourth as the free (or exogenous) variable.

□

(L-Opt-2) Question 2(a) *Es verdadero.* Si $\mathbf{A}$ es simétrica, entonces $\mathbf{A}^\intercal = \mathbf{A}$, por tanto

$$\left(\mathbf{A}^2\right)^\intercal = \left(\mathbf{A}\mathbf{A}\right)^\intercal = \mathbf{A}^\intercal\mathbf{A}^\intercal = \left(\mathbf{A}^\intercal\right)^2 = \mathbf{A}^2,$$

es decir, que $\mathbf{A}^2$ también es simétrica.

□

(L-Opt-2) Question 2(b) *Es verdadero.* Veámoslo:

$$(\mathbf{I} - \mathbf{A})^2 = (\mathbf{I} - \mathbf{A})(\mathbf{I} - \mathbf{A}) = \mathbf{I} - \mathbf{A} - \mathbf{A} + \mathbf{A}^2 = \mathbf{I} - \mathbf{A} - \mathbf{A} + \mathbf{A} = \mathbf{I} - \mathbf{A}$$

Las matrices con esta propiedad se denominan *"matrices idempotentes"*.

□

(L-Opt-2) Question 2(c) *Es falso.* El determinante de una matriz es igual al producto de sus autovalores; si uno de ellos es cero, necesariamente la matriz es singular. En tal caso, sus columnas son linealmente dependientes y es posible encontrar una solución distinta a la trivial ($\boldsymbol{x} = \mathbf{0}$) para dicho sistema homogéneo; así que hay más de una solución y el sistema es necesariamente *indeterminado.*

□

(L-Opt-2) Question 2(d) *Es verdadero.* Por el mismo motivo que antes, si $\mathbf{A}$ es singular, eso implica que el subespacio generado por las columnas de $\mathbf{A}$ (*que llamaremos espacio columna de* $\mathbf{A}$, $\mathcal{C}\left(\mathbf{A}\right)$) es de dimensión menor que m, pero eso quiere decir que existen vectores de $\mathbb{R}^m$ que no pertenecen a $\mathcal{C}\left(\mathbf{A}\right)$. Si $\boldsymbol{b}$ fuera uno de ellos, entonces no existiría una combinación lineal de las columnas de $\mathbf{A}$ igual a $\boldsymbol{b}$, es decir, que $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ será incompatible para dicho $\boldsymbol{b} \notin \mathcal{C}\left(\mathbf{A}\right)$.

□

(L-Opt-2) Question 2(e) *True.* If $\mathbf{Q}$ is orthogonal, then $\mathbf{Q}^\intercal\mathbf{Q} = \mathbf{I}$; implying that the inverse of $\mathbf{Q}$ is its transpose ($\mathbf{Q}^\intercal = \mathbf{Q}^{-1}$), and then $\mathbf{Q}\mathbf{Q}^{-1} = \mathbf{I} = \mathbf{Q}\mathbf{Q}^\intercal$; hence $\mathbf{Q}^{-1}$ also has orthogonal columns (since $\mathbf{Q}\mathbf{Q}^\intercal = \mathbf{I}$ has only zeros outside of the main diagonal) that are of unit norm (since $\mathbf{Q}\ \mathbf{Q}^\intercal$ has only ones on the main diagonal).

□

(L-Opt-2) Question 2(f) *False.* The matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

has the only eigenvalue $\lambda = 1$, but $\mathbf{A}$ is not the identity matrix.

□

(L-Opt-2) Question 3(a) True.

□

(L-Opt-2) Question 3(b) False.

□

(L-Opt-2) Question 3(c) $a_1\boldsymbol{v}_1 + a_2\boldsymbol{v}_2 + \cdots + a_n\boldsymbol{v}_n = \mathbf{0}$ if and only if all coefficients a_j are equal to zero. That is, if the only solution to $\left[\boldsymbol{v}_1, \ldots \boldsymbol{v}_n\right]\boldsymbol{x} = \mathbf{0}$ is a zero vector; that is, if $\mathcal{N}\left(\left[\boldsymbol{v}_1, \ldots \boldsymbol{v}_n\right]\right)$ is $\{\mathbf{0}\}$.

□

(L-Opt-2) Question 3(d) False, because $a_1\mathbf{0} = \mathbf{0}$ even for $a_1 \neq 0$.

□

(L-Opt-2) Question 3(e) we can find n linearly independent eigenvectors.

□

(L-Opt-2) Question 3(f) $\sqrt{7}$

□

(L-Opt-2) Question 3(g) $\boldsymbol{u} \cdot \boldsymbol{v} = 0$

□

(L-Opt-2) Question 4(a) False. $\left|\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right| = 0$

□

(L-Opt-2) Question 4(b) True. $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B}) \neq 0 \Rightarrow \det(\mathbf{A}) \neq 0$ y $\det(\mathbf{B}) \neq 0$.

□

(L-Opt-2) Question 4(c) True. The determinant changes sign, and the product of the eigenvalues is equal to the determinant.

□

(L-Opt-2) Question 4(d) True.

□

(L-Opt-2) Question 4(e) True. In this case $(\mathbf{A} - 5\mathbf{I})$ is of full rank, and $\dim\mathcal{N}\left(\mathbf{A} - 5\mathbf{I}\right) = 0$, therefore there are no eigenvectors associated with the eigenvalue 5.

□

(L-Opt-2) Question 4(f) False. $\left(\begin{smallmatrix} -1 & 1 \\ -1 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 \\ 2 \end{smallmatrix}\right)$ has no solution and $|\mathbf{A}| = 0$.

□

(L-Opt-2) Question 4(g) False. At most it can have three pivotes.

□

(L-Opt-2) Question 4(h) True. $\boldsymbol{b} \in \mathbb{R}^n$ but $\boldsymbol{b} \notin \mathcal{C}\left(\mathbf{C}\right)$; therefore not all vectors in $\mathbb{R}^n$ are in $\mathcal{C}\left(\mathbf{C}\right)$. That is, $\mathrm{rg}\left(\mathbf{C}\right) = \dim\mathcal{C}\left(\mathbf{C}\right) < n = \dim\mathbb{R}^n$.

□

(L-Opt-2) Question 4(i) False. If any eigenvalue is equal to zero, the matrix is not invertible. For example, a zero matrix and square.

□

(L-Opt-2) Question 4(j) True. If it is invertible, it has full rank — n pivots equal to one with zeros above and below... that is, the identity).

□

(L-Opt-2) Question 5(a) $\mathbf{A}$ and $\mathbf{D}$.

□

(L-Opt-2) Question 5(b) Only $\mathbf{B}$ (since it only has two eigenvalues, 0 and 2, necessarily one must be repeated).

□

(L-Opt-2) Question 5(c) $\mathbf{B}$ and $\mathbf{C}$ are not invertible (both have an eigenvalue equal to zero).

□

(L-Opt-2) Question 5(d) $\mathbf{A}$, $\mathbf{C}$ and $\mathbf{D}$ are diagonalizable because they have distinct eigenvalues (in addition, $\mathbf{D}$ is symmetric).

For $\mathbf{B}$, the eigenvalue $\lambda = 2$ is repeated:

$$(\mathbf{B} - 2\mathbf{I}) = \begin{bmatrix} -1 & 1 & 3 \\ 1 & -1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 & 3 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \end{bmatrix}$$

there is only one free column, and therefore, we can only find one direction associated with the repeated eigenvalue $\lambda = 2$. That is, $\mathbf{B}$ *is not diagonalizable.*

□

(L-Opt-2) Question 5(e) For the symmetric matrix $\mathbf{D}$.

□

(L-Opt-2) Question 6(a) Since the matrix is triangular, the eigenvalues are the numbers on the main diagonal: $\lambda_1 = 1$ and $\lambda_2 = 2$.

For $\lambda = 1$

$$(\mathbf{A} - \lambda\mathbf{I}) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

The following are three linearly independent eigenvectors

$$\boldsymbol{x}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}; \quad \boldsymbol{x}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}; \quad \boldsymbol{x}_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

For $\lambda = 2$

$$(\mathbf{A} - 2\lambda\mathbf{I}) = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 1 & 1 & 0 & -1 \end{bmatrix}$$

An eigenvector is

$$\boldsymbol{x}_4 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

□

(L-Opt-2) Question 6(b) Since we have found 4 linearly independent eigenvectors, the matrix is diagonalizable.

□

(L-Opt-2) Question 6(c) This factorization $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{\intercal}$ implies that the matrix $\mathbf{A}$ must be symmetric. Since in this case the matrix is not symmetric, it is not possible to find such a factorization.

□

(L-Opt-2) Question 6(d)

$$|\mathbf{A}^{-1}| = \frac{1}{|\mathbf{A}|} = \frac{1}{\text{product of the eigenvalues of } \mathbf{A}} = \frac{1}{2}.$$

□

(L-Opt-2) Question 7(a) Yes, since its eigenvalues are distinct.

□

(L-Opt-2) Question 7(b) No, it is not. Since $\boldsymbol{v}_3 = -\boldsymbol{v}_1$, we conclude that $\boldsymbol{v}_3$ must be an eigenvector associated with λ_1.

□

(L-Opt-2) Question 7(c)

$$\mathbf{A}(\boldsymbol{v}_1 - \boldsymbol{v}_2) = \mathbf{A}\boldsymbol{v}_1 - \mathbf{A}\boldsymbol{v}_2 = \lambda_1\boldsymbol{v}_1 - \lambda_2\boldsymbol{v}_2 = 1\begin{pmatrix}1\\0\\1\end{pmatrix} - 2\begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}-1\\-2\\-1\end{pmatrix}$$

□

(L-Opt-2) Question 8(a) Since the first two vectors in the solution are the same, the dimension of $\mathcal{N}\left(\mathbf{A}\right)$ is 2. The number of columns must be 4, so the rank of $\mathbf{A}$ is 2..

The last vector tells us that the last column of $\mathbf{A}$ is the zero vector; and the first vector indicates that the first column of $\mathbf{A}$ is the opposite vector of the third column of $\mathbf{A}$. Therefore, one possible solution is:

$$\begin{bmatrix}1 & 0 & -1 & 0\\ 0 & 1 & 0 & 0\end{bmatrix}\begin{pmatrix}x\\y\\z\\t\end{pmatrix} = \begin{pmatrix}0\\0\\0\\0\end{pmatrix} \qquad \Longrightarrow \qquad \begin{cases}x - z &= 0\\ y &= 0\end{cases}.$$

However, this is not the only possible answer; for example, there may be more than two rows (equations) in the system. The requirement is that the rank of $\underset{m\times 4}{\mathbf{A}}$ is 2, that the last column is zero, and that the first is the opposite of the third.

□

(L-Opt-2) Question 8(b) Since the characteristic polynomial of $\mathbf{A}$ has degree 5, we know that $\mathbf{A}$ is a 5×5 matrix. Since 0 is not a root of $p(\cdot)$ and therefore is not an eigenvalue, we conclude that $\mathbf{A}$ is invertible, which implies that its rank is 5.

□

(L-Opt-2) Question 9(a) Since the matrix is invertible, the rank is 3, and the column space $\mathcal{C}\left(\mathbf{A}\right)$ is all of $\mathbb{R}^3$. Therefore, there are no free columns; that is to say, the null space $\mathcal{N}\left(\mathbf{A}\right)$ only contains the zero vector $\mathbf{0}$.

□

(L-Opt-2) Question 9(b) Since $\mathbf{L}$ is $\mathbf{E}^{-1}$ (the inverse of the product of elementary matrices needed to triangularize $\mathbf{A}$), the 5 in the first position of the second row of $\mathbf{L}$ indicates that the first step was *"subtracting five times the first row from the second"*.

Since $\mathbf{U}$ has three pivots, it is of full rank; therefore, it is invertible.

The determinant is $\det(\mathbf{A}) = u_{11} \cdot u_{22} \cdot u_{33}$.

□

(L-Opt-2) Question 9(c) The matrix

$$\mathbf{A} = \begin{bmatrix}0 & 0 & 1\\ 0 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$$

cannot be triangularized without first permuting the rows.

However, by permuting the first row with the third, the matrix can be triangularized. In fact, every invertible matrix admits the following factorization:

$$\mathbf{PA} = \mathbf{LU}.$$

The matrix $\mathbf{A}$ is invertible since, after permuting the rows, three pivots are obtained; that is, the matrix is of full rank.

□